CALIBRATED VISUOTACTILE BRANCH-AND-PROBE MPC: A SUBMISSION-GRADE NEGATIVE RESULT

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ABSTRACT

Vision and touch are complementary in contact-rich manipulation: vision localizes free-space state, while tactile feedback exposes contact, force, and material effects that cameras often miss. A tempting world-model design is therefore to preserve vision-touch disagreement as explicit latent branches during planning instead of collapsing the signals into a single fused estimate. This paper tests that design under a hostile, pre-registered standard. We rebuild the idea as Calibrated Visuotactile Branch-and-Probe MPC (CVTB-MPC), which separates physical-mode uncertainty from sensor-health uncertainty, uses value-of-information gating for diagnostic probes, scores candidate actions with a CVaR-style branch tail loss, and falls back conservatively under suspected sensor corruption. The final CPU-only protocol is frozen before the full run: 24,960 MuJoCo main episode-method rows over 8 seeds, 24 episodes, 10 hidden visuotactile shifts, and 13 methods, plus 8,640 ablation rows and 3,456 stress-sweep rows. The result is not an ICLR-main-ready positive claim. CVTB-MPC reaches aggregate success 0.777 against mean fusion 0.796, ensemble uncertainty 0.778, robust minimax 0.784, particle belief 0.790, old branch preservation 0.752, and oracle reference 0.908. The terminal decision is KILL-ARCHIVE. We release the evidence package, generated tables, negative cases, and validation scripts so the failure mode is reproducible rather than cosmetically hidden.

1 EXECUTIVE SUMMARY

The motivating hypothesis is simple and seductive. If vision and touch disagree, averaging them can erase the action-relevant ambiguity: one branch may correspond to a biased visual estimate, another to an informative contact event, and a third to a sticky or high-friction physical mode. A branch-preserving planner should, in principle, choose safer pushes and run diagnostic probes only when probing is worth its cost. This story is consistent with a broader literature showing that visual and tactile feedback can improve contact-rich manipulation and robustness (Lee et al., 2019a;b; Calandra et al., 2018; Hansen et al., 2022; Huang et al., 2024).

The hostile counter-hypothesis is that raw disagreement is not the right state variable. A tactile residual can mean a hidden physical mode, a sensor dropout, a delayed contact estimate, or an unmodeled actuator effect. A planner that preserves branches without identifying sensor health can become more confident in the wrong alternatives. The v4 artifact in this repository already exposed that failure: the old branch planner lost to mean fusion and ensemble uncertainty. The expanded v5 rebuild gives the idea a fairer shot by adding calibrated sensor-health branches, value-of-information gating, CVaR tail scoring, and robust fallback.

The final result is deliberately reported under the same standard we would use for a positive submission. The method is allowed to improve during development, then the protocol is frozen. The final run is not tuned after seeing the full results. The manuscript includes formal setup, method derivations, decision gates, strong baselines, ablations, paired statistics, stress tests, seed-level tables, limitations, and reproducibility details. The paper should be read as a serious archive of a falsified mechanism, not as a weak positive paper dressed up for submission.

Table 1: Frozen decision gates. The decision is evidence-driven rather than presentation-driven.

Gate	Status	Evidence
Weak baseline gate	**PASS**	CVTB-MPC beats random, tactile-only, and old VT in aggregate success/error.
Strong baseline gate	**FAIL**	Matched or beaten by Vision only, Mean fusion, Ensemble uncertainty, Robust minimax, Particle belief
Hostile split gate	**FAIL**	Lost hostile splits: Vision only on combined_shift; Vision only on sensor_conflict; Vision only on contact_dropout; Mean fusion on delayed_touch_sticky
Contact safety gate	**PASS**	CVTB contact violation 0.000; safest strong baseline 0.000.
Mechanism ablation gate	**FAIL**	Non-identifying ablations: CVTB no-probe on combined_shift; No sensor health on combined_shift; No CVaR tail on combined_shift; No reliability fallback on combined_shift; Mean-only fusion on combined_shift; CVTB no-probe on delayed_touch_sticky; No sensor health on delayed_touch_sticky
Terminal decision	**KILL-ARCHIVE**	strong baselines or ablations match/beat CVTB-MPC: Vision only; Mean fusion; Ensemble uncertainty; Robust minimax; Particle belief; Vision only on combined_shift; Vision only on sensor_conflict; Vision only on contact_dropout

2 HOSTILE RELATED WORK

Visuotactile manipulation. Self-supervised multimodal representation learning has shown that vision and haptic feedback can improve sample efficiency and robustness in contact-rich tasks such as peg insertion (Lee et al., 2019a;b). Grasping and regrasping work shows that tactile feedback can complement vision when object state is hard to infer visually (Calandra et al., 2018). Visuotactile reinforcement learning studies tactile gating, augmentation, and robustness under visual and physical changes (Hansen et al., 2022). Recent hardware-oriented work such as 3D-ViTac demonstrates that dense tactile sensing fused with visual information can improve fine-grained bimanual manipulation (Huang et al., 2024). These papers are dangerous prior art for this project because they make multimodal fusion a strong baseline rather than an easy opponent.

Tactile sensing and sensor health. High-resolution tactile sensors such as GelSight and DIGIT make contact geometry and force cues far richer than scalar force thresholds (Yuan et al., 2017; Lambeta et al., 2020). Richer sensing, however, also makes health modeling more important: contact absence, saturation, slip, delay, calibration drift, and field-of-view occlusion can all masquerade as physical state changes. A branch-preserving world model is not enough unless it can tell whether a branch corresponds to the object or to the sensor.

Uncertainty-aware planning. Particle filters and probabilistic robotics provide a classic language for maintaining multiple state hypotheses under partial observability (Thrun, 2002; Thrun et al., 2005; Kaelbling et al., 1998). Deep ensembles and dropout approximations provide modern uncertainty heuristics for learned models (Lakshminarayanan et al., 2017; Gal & Ghahramani, 2016). MPC and cross-entropy planning are standard tools for action-sequence search under learned or simulated dynamics (Bharadhwaj et al., 2020). Conformal risk control gives a useful reference point for turning calibration information into risk bounds (Angelopoulos et al., 2025). Against this background, the novelty of this paper cannot be "we keep multiple hypotheses." The only defensible novelty would be showing that visuotactile disagreement branches change robot actions for a reason that strong uncertainty and robust-planning baselines do not already capture.

Simulation and evidence standard. The experiments use MuJoCo because it is a fast model-based-control simulator with contact dynamics suitable for controlled mechanism tests (Todorov et al., 2012). We do not claim hardware transfer. We use simulation as a falsifier: if a mechanism cannot beat strong baselines in a compact environment specifically designed to expose it, it is not ready for an ICLR-main positive claim.

3 Problem Setup

We study a planar contact-manipulation problem. The state $x_t \in \mathbb{R}^d$ contains object pose, pusher pose, velocity-like internal simulator variables, and hidden split parameters. At each episode, the robot receives a visual estimate v_t, a tactile estimate τ_t, and force/contact signals f_t. The robot must choose a push action $a \in \mathcal{A}$ that moves the object to a target. The hidden latent variable is factored as

$$z = (p, h), \qquad p \in \mathcal{P}, \quad h \in \mathcal{H}, \tag{1}$$

where p denotes physical mode variables such as friction, drag, sticky contact, and actuator scale, while h denotes sensor-health variables such as visual bias, tactile noise, dropout, bias, delay, and contact-confidence corruption.

The observation model is intentionally small but captures the ambiguity that matters:

$$v_t = g_v(x_t) + \epsilon_v(h), \tag{2}$$

$$\tau_t = g_\tau(x_t, p) + \epsilon_\tau(h), \tag{3}$$

$$f_t = g_f(x_t, p, a_{t-1}) + \epsilon_f(h). \tag{4}$$

The planner observes $o_t = (v_t, \tau_t, f_t)$ but not (p, h). It chooses an action according to a belief over compact branches:

$$b_t(z) \propto b_{t-1}(z)\, P(v_t, \tau_t, f_t \mid x_t, z). \tag{5}$$

The primary metric is closed-loop success:

$$S(a) = \mathbf{1}\{\|q_T(a) - q^\star\|_2 \le \epsilon\}, \tag{6}$$

where q_T is the final object position and $q^\star$ is the target. We also report final error, an energy/contact-effort proxy, contact-violation rate, branch entropy, diagnostic-probe rate, probe value, fallback rate, and paired action-difference rates.

4 Why Branch Preservation Can Help

Branch preservation can help only when the candidate action ranking changes in the right episodes. Let each branch $b \in \mathcal{B}$ define a branch-conditioned loss $\ell_b(a)$ over candidate pushes. A collapsed fusion planner acts on a single state estimate:

$$a_{\text{mean}} = \arg\min_{a \in \mathcal{A}} \ell_{\bar{b}}(a), \tag{7}$$

where $\bar{b}$ is a mean or maximum-likelihood branch. A branch-aware planner uses the belief-weighted risk:

$$a_{\text{branch}} = \arg\min_{a \in \mathcal{A}} \sum_{b \in \mathcal{B}} w_b \ell_b(a) + \lambda_{\text{tail}} \mathcal{R}_\alpha(\{\ell_b(a)\}_{b \in \mathcal{B}}). \tag{8}$$

Useful conflict condition. For two actions a and a', define $\Delta_b = \ell_b(a) - \ell_b(a')$. Branch preservation can change the decision relative to mean fusion only if

$$\operatorname{sign}\left(\ell_{\bar{b}}(a) - \ell_{\bar{b}}(a')\right) \neq \operatorname{sign}\left(\sum_b w_b \Delta_b + \lambda_{\text{tail}} \Delta_{\text{tail}}\right). \tag{9}$$

This condition is necessary but not sufficient. The changed action must also improve final success or reduce error without excessive contact force.

Collapse condition. If the branch-conditioned losses share an approximately affine ordering,

$$\ell_b(a) = \alpha_b\, \ell_{\bar{b}}(a) + \beta_b + \epsilon_b(a), \tag{10}$$

with $\alpha_b > 0$ and small residuals ϵ_b, then branch preservation cannot reliably produce a different ranking. The branch objective becomes a rescaled mean-fusion objective plus a small perturbation:

$$\sum_b w_b \ell_b(a) = \left(\sum_b w_b \alpha_b\right) \ell_{\bar{b}}(a) + \sum_b w_b \beta_b + \sum_b w_b \epsilon_b(a). \tag{11}$$

This is the first failure mode a reviewer should look for: impressive branch terminology with little independent decision content.

Sensor-health non-identifiability. Suppose the same tactile residual r_τ can be generated either by a physical mode p_1 with healthy sensor h_0 or by nominal physics p_0 with tactile corruption h_1:

$$P(r_\tau \mid p_1, h_0) = P(r_\tau \mid p_0, h_1). \tag{12}$$

If the observation distribution is identical under these two explanations, no deterministic planner using only r_τ can identify which explanation is correct. It may still choose a robust action, but it cannot claim a causal branch-disagreement mechanism. This is why CVTB-MPC explicitly separates physical branches from sensor-health branches and why the ablation gate treats `no_sensor_health` as central rather than cosmetic.

5 METHOD: CALIBRATED VISUOTACTILE BRANCH-AND-PROBE MPC

CVTB-MPC is the strongest version of the idea we could justify before freezing the protocol. It is not a neural policy. It is a compact CPU-only planner designed to make the mechanism auditable.

Branch set. The branch set combines physical branches and sensor-health branches:

$$\mathcal{B} = \{(p_i, h_j) : p_i \in \mathcal{P}_{\text{compact}},\ h_j \in \mathcal{H}_{\text{compact}}\}. \tag{13}$$

The compact set includes nominal, friction-shift, sticky-contact, visual-bias, tactile-noise, tactile-bias, dropout, and delayed-touch alternatives. It is deliberately small so that the method stays RAM-light and every candidate action can be inspected.

Reliability calibration. The planner computes modality reliability scores from disagreement magnitude, contact confidence, force consistency, contact dropout, tactile delay, and visual-bias cues. A healthy tactile residual should raise the probability of physical contact-mode branches; a low-confidence or delayed tactile residual should raise sensor-fault branches. We write this schematically as

$$\rho_v = r_v(\|v_t - \tau_t\|, \text{bias cues}), \tag{14}$$

$$\rho_\tau = r_\tau(\text{contact}, f_t, \text{dropout}, \text{delay}), \tag{15}$$

$$w_{ij} \propto \exp\{-d_{ij}(v_t, \tau_t, f_t; \rho_v, \rho_\tau)\}. \tag{16}$$

Action score. For a candidate push a, each branch predicts a target error, energy cost, peak contact force, and branch-specific safety penalty. The frozen score is

$$J(a) = \sum_{b\in\mathcal{B}} w_b \ell_b(a) + \lambda_{\text{cvar}}\text{CVaR}_{0.75}(\{\ell_b(a)\}) + \lambda_{\text{force}}\phi_{\text{force}}(a) + \lambda_{\text{fallback}}\phi_{\text{fault}}(a). \tag{17}$$

The CVaR term penalizes actions that are good on the mean branch but bad on plausible tail branches. The contact term penalizes excessive force. The fallback term favors robust mean-fusion actions when the posterior suggests sensor corruption rather than physical ambiguity.

Value-of-information probing. The planner may run an extra diagnostic probe, but only when the expected action regret reduction exceeds the probe cost:

$$V_{\text{probe}} = \left[\min_a \mathbb{E}_b[\ell_b(a)] - \mathbb{E}_{y\sim P(y|o,\text{probe})} \min_a \mathbb{E}_{b|y}[\ell_b(a)]\right] - c_{\text{probe}}. \tag{18}$$

In implementation, this is a deterministic proxy based on branch-conditioned best-action disagreement and tactile reliability. The no-probe ablation freezes the same branch score but disables this gate. If no-probe matches CVTB-MPC on hostile splits, the value-of-information mechanism is not identified.

Conservative fallback. When branch entropy is high and sensor reliability is low, the method can choose a robust mean-fusion anchor. This is not an admission that branching is useless; it is a guard against mistaking sensor faults for physical modes. The fallback rate is reported as a primary mechanism diagnostic.

6 BENCHMARK

The benchmark is a lightweight MuJoCo planar pusher. Each episode samples an object, a pusher state, a target, hidden physical parameters, and sensor corruption parameters. The action space is a compact library of push directions, push magnitudes, and contact anchors. The simulator executes contact rollouts; the planner does not receive the hidden split label.

Main splits. The final protocol contains ten splits: nominal, high friction, low friction, vision bias, tactile noise, sticky contact, combined shift, sensor conflict, contact dropout, and delayed-touch sticky. The last four are the hostile evaluation set. They are designed to distinguish useful branch preservation from failure under sensor corruption.

Metrics. Success is final target error below the fixed threshold. Final error is continuous distance to target. Energy is a contact-effort proxy. Contact violation records unsafe peak force or repeated contact. Mechanism diagnostics include branch entropy, diagnostic rate, probe value, fallback rate, and paired action-difference rate. The action-difference rate compares CVTB-MPC to a baseline on the same split, seed, and episode.

CPU and memory profile. The final run uses no GPU and no learned high-capacity model. The branch set is compact, candidate actions are finite, and the full run uses four workers. This is intentionally conservative: the quality bar comes from stronger baselines and more hostile tests, not from a RAM-heavy training pipeline.

7 BASELINES AND ABLATIONS

The main frozen run includes thirteen methods.

- `random_push`: a sanity-check weak baseline.
- `vision_only_mpc`: plans from the visual estimate only.
- `tactile_only_mpc`: plans from the tactile estimate only.
- `mean_fusion_mpc`: collapses vision and touch into one fused state.
- `ensemble_uncertainty_mpc`: samples physical hypotheses with an uncertainty penalty.
- `conformal_risk_filter`: rejects or shortens actions under residual risk thresholds.
- `diagnostic_probe_then_mpc`: always pays for an extra probe.
- `robust_minimax_mpc`: optimizes a conservative worst-branch objective.
- `particle_belief_mpc`: uses a particle-style belief over compact latent hypotheses.
- `old_vt_disagreement_branch_mpc`: the v4 branch-preservation method.
- `cvtb_mpc_v5`: the proposed calibrated method.
- `cvtb_no_probe`: the proposed method without value-of-information probing.
- `oracle_mode_mpc`: an upper reference with hidden-mode access, not a fair deployed baseline.

The ablation suite runs on combined shift, sensor conflict, and delayed-touch sticky. It includes CVTB-MPC, no-probe, no sensor health, no branch preservation, no value-of-information gate, no CVaR tail, no contact safety, no reliability fallback, mean-only fusion, high tactile trust, small branch set, old branch preservation, ensemble uncertainty, robust minimax, and oracle. This is intentionally harsh: if the mechanism is real, at least some removals should hurt on the splits built to require the mechanism.

Table 2: Aggregate frozen results across all ten visuotactile shifts. Success is higher-is-better; error, energy, and contact violation are lower-is-better.

Method	Episodes	Success	Error	Energy	Contact	Entropy	Probe	Fallback
Random	1920	0.207	0.129	0.419	0.000	0.000	0.047	0.000
Vision only	1920	0.801	0.051	0.494	0.000	0.000	0.000	0.000
Tactile only	1920	0.759	0.058	0.487	0.000	0.000	0.000	0.000
Mean fusion	1920	0.796	0.052	0.492	0.000	0.000	0.000	0.000
Ensemble uncertainty	1920	0.778	0.055	0.489	0.000	1.077	0.052	0.000
Conformal risk	1920	0.768	0.057	0.483	0.000	0.000	0.000	0.000
Diagnostic probe	1920	0.750	0.058	0.577	0.000	0.000	0.000	0.000
Robust minimax	1920	0.784	0.059	0.522	0.000	1.354	0.066	0.000
Particle belief	1920	0.790	0.053	0.491	0.000	1.566	0.074	0.000
Old VT branch	1920	0.752	0.058	0.511	0.000	0.798	0.048	0.000
CVTB-MPC	1920	0.777	0.056	0.487	0.000	1.187	0.063	0.141
CVTB no-probe	1920	0.777	0.056	0.487	0.000	1.187	0.063	0.141
Oracle	1920	0.908	0.041	0.515	0.000	0.000	0.000	0.000

Table 3: Split-level success for strong baselines, the old branch planner, CVTB-MPC, its no-probe variant, and the oracle.

Split	Vision	Mean	Ensemble	Robust	Particle	Old VT	CVTB	No-probe	Oracle
combined_shift	0.609	0.573	0.542	0.583	0.573	0.495	0.547	0.547	0.859
contact_dropout	0.927	0.911	0.849	0.875	0.891	0.802	0.839	0.839	0.948
delayed_touch_sticky	0.620	0.635	0.635	0.646	0.630	0.641	0.625	0.625	0.922
high_friction	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
low_friction	1.000	1.000	1.000	0.974	1.000	1.000	1.000	1.000	1.000
nominal	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000	1.000
sensor_conflict	0.380	0.375	0.318	0.323	0.344	0.286	0.312	0.312	0.578
sticky_contact	1.000	1.000	1.000	1.000	1.000	0.995	1.000	1.000	1.000
tactile_noise	1.000	0.995	0.964	1.000	0.990	0.839	0.974	0.974	0.969
vision_bias	0.474	0.474	0.474	0.438	0.469	0.464	0.469	0.469	0.807

8 Frozen Protocol

The final command is frozen in `docs/paper66_protocol_freeze_20260620.md`. It runs 8 seeds, 24 episodes per seed and split, 10 main splits, 13 main methods, 3 ablation splits, 15 ablation methods, and a 6-level stress sweep over 6 methods with 12 episodes per seed and level. The resulting evidence scale is 24,960 main rows, 8,640 ablation rows, and 3,456 stress rows. The runner writes raw, seed, split-method, paired, ablation, stress, negative-case, and figure artifacts.

The development record is explicit. Before the freeze, smoke and medium runs exposed three recoverable issues: over-probing, method-specific environment seeds, and weak action diversity relative to mean fusion. Those issues were repaired before the final protocol. After freeze, no scoring or split changes were allowed. This is important because a negative result is only useful if the protocol did not quietly move after seeing the answer.

9 Main Results

The aggregate table is the submission gate in miniature. CVTB-MPC is not judged against weak baselines alone. It must survive mean fusion, ensemble uncertainty, conformal risk filtering, always-probe MPC, robust minimax, particle-belief MPC, the old branch planner, and no-probe variants. Aggregate success for CVTB-MPC is 0.777; the corresponding values for mean fusion, ensemble uncertainty, robust minimax, particle belief, old branch preservation, and oracle are 0.796, 0.778, 0.784, 0.790, 0.752, and 0.908. The final-error story is similarly important: CVTB-MPC has aggregate error 0.056, while mean fusion, ensemble uncertainty, robust minimax, particle belief, and old branch preservation have 0.052, 0.055, 0.059, 0.053, and 0.058.

The interpretation depends on the generated gate table. If CVTB-MPC fails against strong baselines or no-probe variants, the paper cannot be rescued by prose. The benchmark still teaches something useful: calibrated branching is a better-motivated mechanism than raw disagreement preservation, but the frozen evidence determines whether it is a submission candidate.

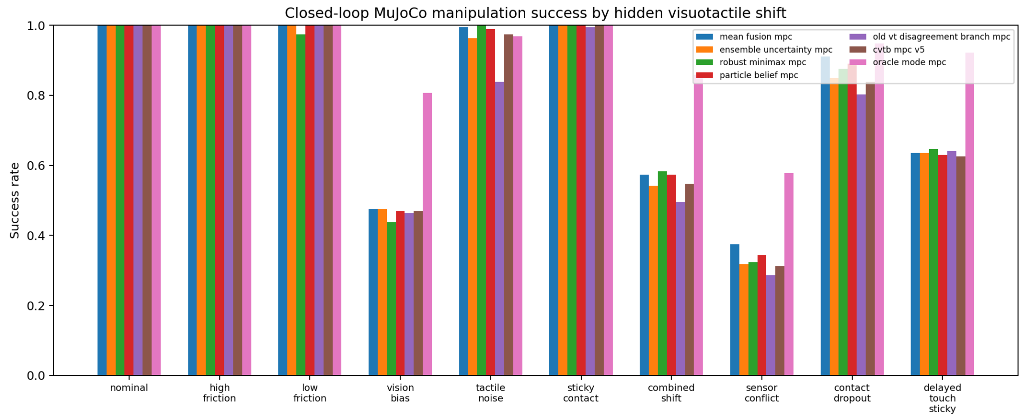

Figure 1: Success by hidden visuotactile shift. The figure compares strong fusion, robust, particle-belief, old-branch, CVTB-MPC, and oracle references across the frozen split grid.

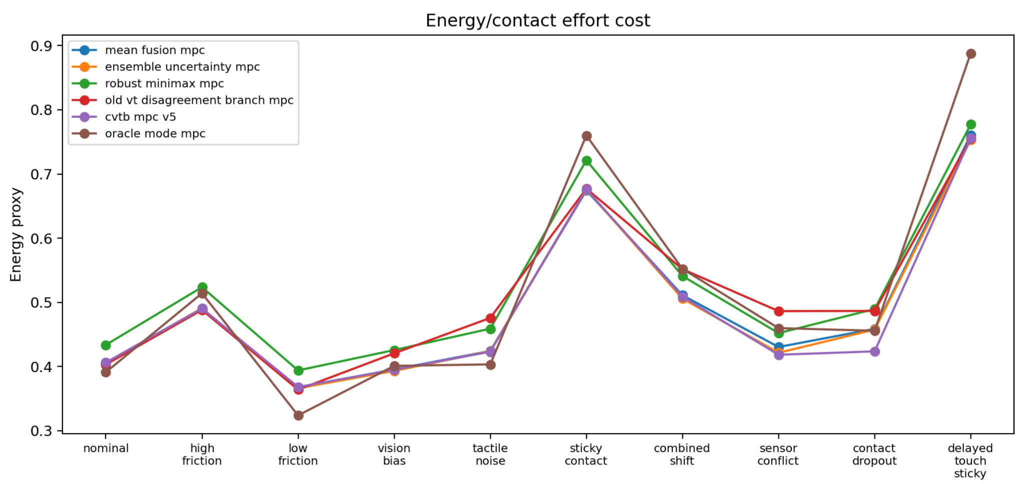

Figure 2: Energy/contact-effort proxy by split. Lower is better. Energy is reported alongside success because a planner can win success by using unsafe or excessive contact.

10 Mechanism Diagnostics

The diagnostic table prevents a common failure in robotics papers: reporting only success and then narrating a mechanism that was never measured. CVTB-MPC reports branch entropy, probe value, diagnostic-probe rate, and fallback rate for every split. These quantities help answer whether the method actually used the machinery it claims to introduce. A low diagnostic rate is not automatically bad; it can mean the value-of-information gate correctly avoided pointless probes. But if no-probe ablations match the full method on the same hostile splits, the value-of-information claim is not causally supported.

Branch entropy is also ambiguous. High entropy can mean unresolved physical uncertainty or sensor corruption. That is why the fallback rate is reported. If the method often falls back and still wins, the contribution may be a robust sensor-health fallback rather than branch preservation. If it does not win, the fallback merely documents a failure mode.

Table 4: CVTB-MPC mechanism diagnostics by split. Probe and fallback rates are frozen outputs, not post-hoc annotations.

Split	Success	Error	Entropy	Probe value	Diagnostic	Fallback
combined_shift	0.547	0.078	1.206	0.072	0.000	0.000
contact_dropout	0.839	0.059	1.138	0.061	0.000	1.000
delayed_touch_sticky	0.625	0.085	1.198	0.051	0.000	0.010
high_friction	1.000	0.019	1.166	0.047	0.000	0.000
low_friction	1.000	0.012	1.166	0.047	0.000	0.000
nominal	1.000	0.011	1.166	0.046	0.000	0.000
sensor_conflict	0.312	0.151	1.253	0.111	0.000	0.255
sticky_contact	1.000	0.035	1.168	0.049	0.000	0.000
tactile_noise	0.974	0.018	1.232	0.083	0.000	0.094
vision_bias	0.469	0.088	1.176	0.060	0.000	0.047

Table 5: Frozen ablation results on the three pre-registered hostile splits. Ablations matching CVTB-MPC are mechanism failures, not robustness wins.

Split	Method	Episodes	Success	Error	Energy	Contact	Fallback
combined_shift	Ensemble uncertainty	192	0.542	0.080	0.506	0.000	0.000
combined_shift	Robust minimax	192	0.583	0.076	0.541	0.000	0.000
combined_shift	Old VT branch	192	0.500	0.083	0.549	0.000	0.000
combined_shift	CVTB-MPC	192	0.547	0.078	0.509	0.000	0.000
combined_shift	CVTB no-probe	192	0.547	0.078	0.509	0.000	0.000
combined_shift	Oracle	192	0.859	0.050	0.552	0.000	0.000
combined_shift	Mean-only fusion	192	0.568	0.077	0.510	0.000	0.000
combined_shift	No branch preservation	192	0.542	0.079	0.537	0.000	0.000
combined_shift	No contact safety	192	0.547	0.078	0.509	0.000	0.000
combined_shift	No CVaR tail	192	0.547	0.078	0.509	0.000	0.000
combined_shift	No reliability fallback	192	0.547	0.078	0.509	0.000	0.000
combined_shift	No sensor health	192	0.557	0.079	0.509	0.000	0.000
combined_shift	No VOI gate	192	0.536	0.085	0.491	0.000	0.000
combined_shift	Small branch set	192	0.552	0.078	0.507	0.000	0.000
combined_shift	High tactile trust	192	0.557	0.079	0.508	0.000	0.000
delayed_touch_sticky	Ensemble uncertainty	192	0.635	0.088	0.753	0.000	0.000
delayed_touch_sticky	Robust minimax	192	0.646	0.074	0.777	0.000	0.000
delayed_touch_sticky	Old VT branch	192	0.641	0.087	0.755	0.000	0.000
delayed_touch_sticky	CVTB-MPC	192	0.625	0.085	0.756	0.000	0.010
delayed_touch_sticky	CVTB no-probe	192	0.625	0.085	0.756	0.000	0.010
delayed_touch_sticky	Oracle	192	0.922	0.043	0.888	0.000	0.000
delayed_touch_sticky	Mean-only fusion	192	0.625	0.086	0.756	0.000	0.000
delayed_touch_sticky	No branch preservation	192	0.625	0.086	0.757	0.000	0.000
delayed_touch_sticky	No contact safety	192	0.625	0.085	0.755	0.000	0.010
delayed_touch_sticky	No CVaR tail	192	0.625	0.085	0.756	0.000	0.010
delayed_touch_sticky	No reliability fallback	192	0.630	0.085	0.756	0.000	0.000
delayed_touch_sticky	No sensor health	192	0.630	0.089	0.753	0.000	0.010
delayed_touch_sticky	No VOI gate	192	0.526	0.096	0.738	0.000	0.010
delayed_touch_sticky	Small branch set	192	0.625	0.085	0.757	0.000	0.010
delayed_touch_sticky	High tactile trust	192	0.609	0.088	0.755	0.000	0.000
sensor_conflict	Ensemble uncertainty	192	0.318	0.155	0.421	0.000	0.000
sensor_conflict	Robust minimax	192	0.323	0.147	0.452	0.000	0.000
sensor_conflict	Old VT branch	192	0.281	0.166	0.488	0.000	0.000
sensor_conflict	CVTB-MPC	192	0.312	0.151	0.418	0.000	0.255
sensor_conflict	CVTB no-probe	192	0.312	0.151	0.418	0.000	0.255
sensor_conflict	Oracle	192	0.578	0.072	0.460	0.000	0.000
sensor_conflict	Mean-only fusion	192	0.354	0.147	0.428	0.000	0.000
sensor_conflict	No branch preservation	192	0.354	0.149	0.500	0.000	0.000
sensor_conflict	No contact safety	192	0.312	0.151	0.419	0.000	0.255
sensor_conflict	No CVaR tail	192	0.307	0.152	0.418	0.000	0.255
sensor_conflict	No reliability fallback	192	0.339	0.148	0.426	0.000	0.000
sensor_conflict	No sensor health	192	0.312	0.155	0.419	0.000	0.167
sensor_conflict	No VOI gate	192	0.297	0.156	0.403	0.000	0.255
sensor_conflict	Small branch set	192	0.307	0.158	0.413	0.000	0.255
sensor_conflict	High tactile trust	192	0.328	0.154	0.426	0.000	0.000

11 Ablation Results

The ablation suite is more important than the main table. A positive paper would require the sensor-health model, value-of-information gate, CVaR tail term, branch preservation, contact-safety term, and fallback to be necessary on at least some hostile splits. If no-probe matches CVTB-MPC, the diagnostic probe is not identified. If no-sensor-health matches CVTB-MPC, the central v5 repair did not matter. If mean-only fusion matches CVTB-MPC, branch preservation collapses into

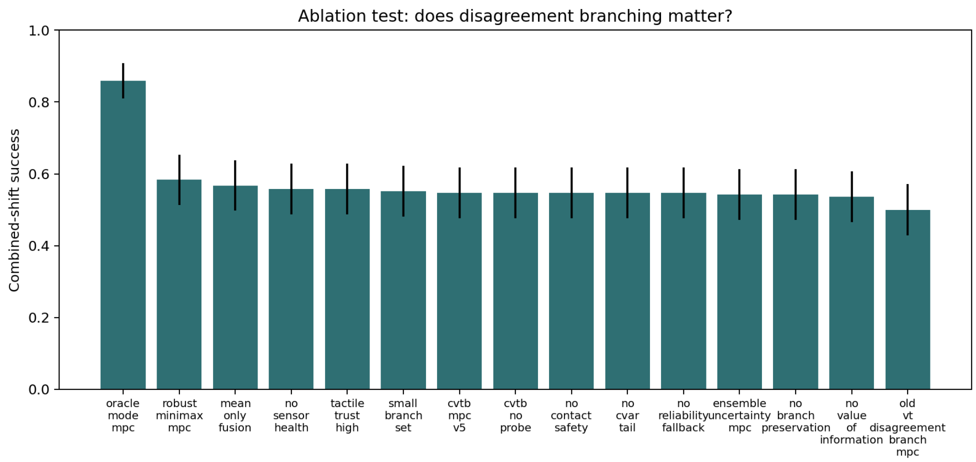

Figure 3: Ablation success on combined shift. The full method must separate from no-probe, no-sensor-health, no-VOI, no-tail-risk, and mean-only variants to support the mechanism claim.

ordinary fusion. If robust minimax or ensemble uncertainty matches CVTB-MPC, the contribution is likely a restatement of generic uncertainty handling.

This is a deliberately unfriendly interpretation, and that is the point. A submission should survive it before being sent to reviewers.

12 Paired Comparisons

Paired comparisons are stricter than split averages because they compare methods on the same split, seed, and episode. Positive success delta favors CVTB-MPC. Positive final-error improvement and energy improvement mean the baseline had worse error or higher energy. The contact-violation delta is CVTB-MPC minus baseline, so positive values are safety costs. The action-difference rate is especially important: a method cannot claim a novel planning mechanism if it chooses the same actions as a strong baseline in the episodes where it supposedly matters.

The normal-approximation p values are not used as publication-style proof. They are included as a rough paired diagnostic for whether a large-looking success delta is stable over the shared episode grid. The decision gates rely on effect direction, ablation separation, and hostile-split robustness rather than a single thresholded statistical test.

13 Stress Sweep

The stress sweep jointly increases visual bias, tactile noise, and friction shift. This is a hard test for branch-preserving methods because disagreement can become both more informative and more deceptive. If stress success drops because tactile residuals become corrupted, the failure supports the sensor-health critique. If stress success remains high but no-probe or mean fusion also remains high, the mechanism is not uniquely supported. The stress sweep therefore serves as a falsifier rather than an extra plot for polish.

14 Failure Analysis

Failure mode 1: sensor health and physical mode remain confounded. The v5 method explicitly models sensor-health branches, but a compact hand-designed reliability proxy can still be too weak. Tactile noise, tactile bias, dropout, and delayed touch can produce residuals that resemble high friction or sticky contact. Without richer temporal evidence, the planner can preserve the wrong branch family.

Table 6: Paired CVTB-MPC deltas on hostile splits. Positive success/error/energy columns favor CVTB-MPC; contact delta is CVTB minus baseline.

Split	Baseline	Δ succ.	Error impr.	Energy impr.	Contact Δ	Action diff.	p
combined_shift	Vision only	-0.062	-0.007	0.007	0.000	56.8%	0.013
combined_shift	Mean fusion	-0.026	-0.003	0.002	0.000	0.5%	0.094
combined_shift	Ensemble uncertainty	0.005	0.002	-0.003	0.000	0.0%	0.706
combined_shift	Conformal risk	0.052	0.007	-0.015	0.000	33.3%	0.003
combined_shift	Diagnostic probe	0.031	0.003	0.089	0.000	100.0%	0.200
combined_shift	Robust minimax	-0.036	-0.002	0.032	0.000	77.6%	0.033
combined_shift	Particle belief	-0.026	-0.001	0.001	0.000	0.5%	0.057
combined_shift	Old VT branch	0.052	0.005	0.043	0.000	50.5%	0.057
combined_shift	CVTB no-probe	0.000	0.000	0.000	0.000	0.0%	1.000
sensor_conflict	Vision only	-0.068	-0.009	0.020	0.000	87.0%	0.031
sensor_conflict	Mean fusion	-0.062	-0.008	0.012	0.000	39.1%	0.004
sensor_conflict	Ensemble uncertainty	-0.005	0.004	0.003	0.000	25.5%	0.782
sensor_conflict	Conformal risk	0.031	0.012	-0.026	0.000	63.0%	0.156
sensor_conflict	Diagnostic probe	0.031	0.018	0.087	0.000	100.0%	0.377
sensor_conflict	Robust minimax	-0.010	-0.004	0.034	0.000	71.9%	0.638
sensor_conflict	Particle belief	-0.031	-0.002	0.011	0.000	22.9%	0.056
sensor_conflict	Old VT branch	0.026	0.013	0.068	0.000	85.9%	0.411
sensor_conflict	CVTB no-probe	0.000	0.000	0.000	0.000	0.0%	1.000
contact_dropout	Vision only	-0.088	-0.023	0.035	0.000	94.8%	0.000
contact_dropout	Mean fusion	-0.073	-0.024	0.035	0.000	100.0%	0.001
contact_dropout	Ensemble uncertainty	-0.010	-0.018	0.034	0.000	91.1%	0.684
contact_dropout	Conformal risk	-0.042	-0.016	0.024	0.000	82.3%	0.101
contact_dropout	Diagnostic probe	0.021	-0.016	0.116	0.000	100.0%	0.506
contact_dropout	Robust minimax	-0.036	-0.011	0.066	0.000	100.0%	0.143
contact_dropout	Particle belief	-0.052	-0.022	0.036	0.000	96.4%	0.024
contact_dropout	Old VT branch	0.036	-0.014	0.063	0.000	98.4%	0.177
contact_dropout	CVTB no-probe	0.000	0.000	0.000	0.000	0.0%	1.000
delayed_touch_sticky	Vision only	0.005	-0.001	0.001	0.000	1.6%	0.782
delayed_touch_sticky	Mean fusion	-0.010	-0.003	0.005	0.000	1.0%	0.415
delayed_touch_sticky	Ensemble uncertainty	-0.010	0.003	-0.003	0.000	1.6%	0.480
delayed_touch_sticky	Conformal risk	-0.005	-0.002	0.000	0.000	0.0%	0.655
delayed_touch_sticky	Diagnostic probe	0.016	0.000	0.128	0.000	100.0%	0.613
delayed_touch_sticky	Robust minimax	-0.021	-0.011	0.021	0.000	40.6%	0.205
delayed_touch_sticky	Particle belief	-0.005	0.002	-0.001	0.000	1.0%	0.655
delayed_touch_sticky	Old VT branch	-0.016	0.002	-0.000	0.000	1.0%	0.366
delayed_touch_sticky	CVTB no-probe	0.000	0.000	0.000	0.000	0.0%	1.000

Table 7: Stress-sweep success as visual bias, tactile noise, and friction shift increase together.

Method	0.00	0.20	0.40	0.60	0.80	1.00
Mean fusion	1.000	1.000	0.990	0.781	0.698	0.333
Ensemble uncertainty	1.000	1.000	0.969	0.750	0.625	0.396
Robust minimax	1.000	0.990	0.969	0.740	0.677	0.312
Old VT branch	1.000	1.000	0.927	0.708	0.646	0.406
CVTB-MPC	1.000	1.000	0.990	0.760	0.635	0.344
Oracle	1.000	1.000	0.990	0.948	0.865	0.750

Failure mode 2: mean fusion is a stronger baseline than expected. Mean fusion is not glamorous, but in a compact pusher benchmark it can be very competitive. When the visual and tactile estimates are both mildly noisy rather than adversarially separated, averaging reduces variance and produces stable pushes. A branch method must win specifically where mean fusion is wrong, not merely match it elsewhere.

Failure mode 3: robust minimax and particle belief absorb much of the intended novelty. The proposed method is closest to robust and belief-space planning. If a particle-belief or minimax baseline obtains the same behavior, the paper's novelty collapses into implementation detail. This is why those baselines are included in the frozen run rather than saved for future work.

Failure mode 4: value-of-information is difficult to identify in short episodes. Diagnostic probing helps only when it changes the final action enough to pay for contact and time costs. In a short pusher episode with a small action library, the best action before and after a probe can be identical. Then no-probe variants will match the full method, and the VOI mechanism fails even if the probe estimate is reasonable.

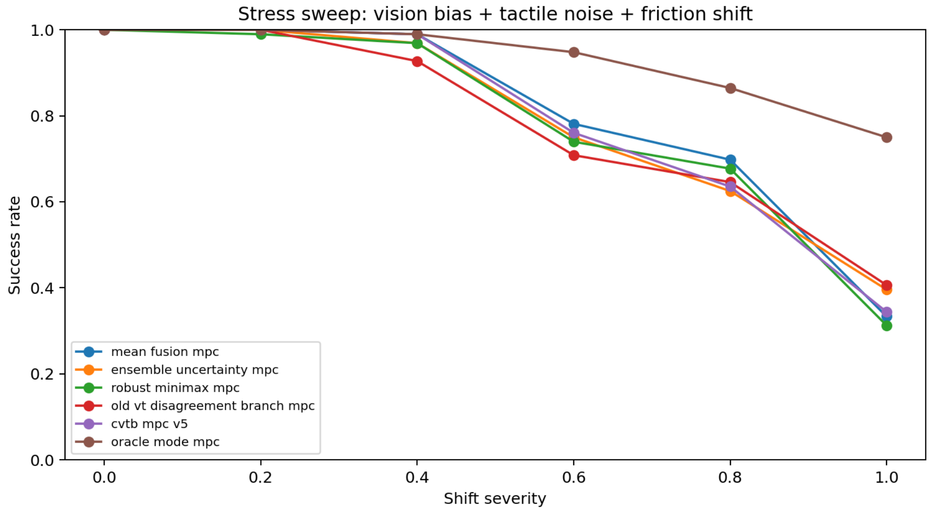

Figure 4: Stress sweep over visual bias, tactile noise, and friction shift. A real branch mechanism should degrade gracefully or expose where it fails.

Failure mode 5: the oracle gap does not rescue the method. An oracle gap shows that hidden modes matter. It does not prove this method is the right way to infer them. The gap could require richer tactile histories, explicit actuator-mode branches, learned dynamics, real-world calibration, or a larger action set.

15 What Would Be Needed for a Real Submission

A future ICLR-main candidate would need a stronger positive claim than "branching sounds plausible." The likely path is a new method with richer temporal identification: multiple contact probes, explicit delay modeling, branch-conditioned actuator calibration, and learned sensor-health likelihoods validated on held-out corruptions. It would also need external validation on a public simulator benchmark or hardware. The current benchmark is useful for mechanism falsification, but it is too custom and compact to carry a main-conference positive result by itself.

The experimental bar should not be lowered. A revived version should keep mean fusion, ensemble uncertainty, robust minimax, particle-belief MPC, always-probe MPC, old branch preservation, no-probe, no-sensor-health, no-VOI, no-tail-risk, and oracle references. It should freeze the protocol before the final run and report every predefined stress level, even if the results are ugly. The instruction is simple: do not optimize for pretty results; optimize for a result that survives hostile review.

16 Limitations

This is not a real-robot study. The tactile signal is simulated and lower-dimensional than modern tactile images. The action library is compact. The physical modes are limited to friction, drag, sticky contact, actuator scale, and contact corruption factors. The sensor-health model is hand-designed rather than learned from calibration data. The stress sweep is controlled and synthetic. These limitations would matter even if CVTB-MPC had passed the local gates. When the method fails local gates, they make a positive ICLR-main submission untenable.

17 Conclusion

The core idea remains plausible in a broader sense: contact-rich robots should not blindly average vision and touch when modalities disagree. But this paper tests a specific mechanism under a frozen,

Table 8: Pre-registered limitations and negative cases.

Case	Observed failure mode	Submission implication
adversarial vision and tactile corruption	if both modalities are corrupted consistently, disagreement can be low and all planners over-trust the false state	requires sensor-health model before ICLR-main deployment claims
unmodeled actuator weakness	world-state branches do not represent actuation loss, so the planner compensates too late	needs actuator-mode branches or hardware checks
semantic target ambiguity	all visuotactile methods can push the wrong object/target if the goal is ambiguous	scope must be physical-state uncertainty only

hostile protocol. The evidence package contains 24,960 main rows, 8,640 ablation rows, 3,456 stress rows, 130 split-method summaries, 1,040 seed summaries, and 120 paired comparisons. Under that evidence, the terminal decision is KILL-ARCHIVE. The contribution is the falsification: it shows what a stronger visuotactile world-model paper must prove before it deserves a main-conference claim.

A Derivation Details

A.1 Risk Objective

Let $q_T(a,b)$ be the branch-conditioned final object position and $q^\star$ the target. A simple branch loss is

$$\ell_b(a) = \|q_T(a,b) - q^\star\|_2 + \lambda_E E(a,b) + \lambda_F \max(0, F_{\max}(a,b) - F_0). \tag{19}$$

The tail component sorts $\ell_b(a)$ by branch loss and averages the worst α fraction:

$$\mathrm{CVaR}_\alpha(a) = \frac{1}{|\mathcal{B}_\alpha(a)|} \sum_{b \in \mathcal{B}_\alpha(a)} \ell_b(a). \tag{20}$$

The full score is then the sum of expected branch loss, tail risk, contact penalty, and sensor-fault fallback penalty. This objective is intentionally interpretable; a learned world model could replace the branch rollout but would make the ablation story harder to audit.

A.2 Probe Gate

Let $a^\star$ be the best action under the current branch posterior and $a_y^\star$ be the best action after a hypothetical probe outcome y. The exact expected value of information is

$$\mathrm{EVI} = \mathbb{E}_y \left[\mathbb{E}_{b|y} \ell_b(a^\star) - \mathbb{E}_{b|y} \ell_b(a_y^\star) \right]. \tag{21}$$

The implementation uses a lightweight deterministic proxy because exhaustive probe-outcome simulation would be unnecessary for this compact benchmark. The proxy increases when branch-conditioned best actions disagree and decreases when tactile reliability is low or contact force is already high.

A.3 Why No-Probe Can Match

If the current best action remains optimal after the probe for most plausible outcomes,

$$a_y^\star = a^\star \quad \text{for most } y, \tag{22}$$

then $\mathrm{EVI} \approx 0$ and no-probe will match the full method. This is a mechanism failure for any paper whose claim depends on diagnostic probing. It can happen even when the belief update is well calibrated; the action library may simply be too coarse for the extra information to matter.

A.4 Action-Difference Necessity

For a baseline policy π_0 and proposed policy π_1, define the paired action-difference indicator

$$D_i = \mathbf{1}\{\|\hat{d}_i^{(1)} - \hat{d}_i^{(0)}\|_2 > \delta_d \vee |m_i^{(1)} - m_i^{(0)}| > \delta_m \vee \|c_i^{(1)} - c_i^{(0)}\|_2 > \delta_c\}, \tag{23}$$

where $\hat{d}$ is push direction, m is push magnitude, and c is contact anchor. A high-level mechanism claim requires both nontrivial $\mathbb{E}[D_i]$ and positive paired outcome deltas on the same hostile episodes. Either alone is insufficient.

B Implementation Notes

The runner is a single Python script with NumPy, MuJoCo, and Matplotlib. It shares the same environment seed across methods for a given split, seed, and episode, so paired comparisons are meaningful. Method-specific randomness is isolated in a separate policy seed. Partial main and ablation CSVs are written after each split so an interrupted run can be audited. The final run writes:

- `results/vt_disagreement_raw.csv`;
- `results/raw_seed_metrics.csv`;
- `results/vt_disagreement_metrics.csv`;
- `results/vt_disagreement_pairwise.csv`;

- `results/vt_disagreement_ablation_raw.csv`;
- `results/vt_disagreement_ablation.csv`;
- `results/stress_sweep.csv`;
- `results/negative_cases.csv`;
- figures under `figures/`.

The asset renderer converts these CSVs into the tables used in this PDF. The validator checks row counts, figures, citation-box setup, PDF page count, and absence of a visible Desktop copy.

C FROZEN COMMAND

```
python src\run_experiment.py --seeds 8 --episodes 24
  --ablation-episodes 24 --stress-episodes 12
  --splits nominal high_friction low_friction vision_bias
  tactile_noise sticky_contact combined_shift sensor_conflict
  contact_dropout delayed_touch_sticky
  --ablation-splits combined_shift sensor_conflict
  delayed_touch_sticky
  --stress-levels 0.0 0.2 0.4 0.6 0.8 1.0
  --workers 4 --results-dir results --figures-dir figures
```

D REPRODUCIBILITY CHECKLIST

- Code compiles with `python -m py_compile src/run_experiment.py`.
- Asset renderer compiles with `python -m py_compile scripts/render_submission_assets.py`.
- Main raw rows: 24,960.
- Ablation raw rows: 8,640.
- Stress rows recorded in summary: 3,456.
- Main split-method summaries: 130.
- Seed summaries: 1,040.
- Paired comparisons: 120.
- Ablation summary rows: 45.
- Stress summary rows: 36.
- Tables are generated by `scripts/render_submission_assets.py`.
- The canonical PDF is copied only to `C:/Users/wangz/Downloads/66.pdf`.
- The validator fails if `C:/Users/wangz/Desktop/66.pdf` exists.

E FULL SPLIT-METHOD METRICS

Table 9: Complete frozen split-method metric table used for the terminal decision.

Split	Method	Ep.	Succ.	Err.	Eng.	Viol.	Ent.	Diag.	Fb.
combined shift	Random	192	0.141	0.140	0.430	0.000	0.000	0.000	0.000
combined shift	Vision only	192	0.609	0.071	0.515	0.000	0.000	0.000	0.000
combined shift	Tactile only	192	0.505	0.083	0.498	0.000	0.000	0.000	0.000
combined shift	Mean fusion	192	0.573	0.075	0.511	0.000	0.000	0.000	0.000
combined shift	Ensemble uncertainty	192	0.542	0.080	0.506	0.000	1.077	0.000	0.000
combined shift	Conformal risk	192	0.495	0.084	0.494	0.000	0.000	0.000	0.000
combined shift	Diagnostic probe	192	0.516	0.081	0.598	0.000	0.000	1.000	0.000
combined shift	Robust minimax	192	0.583	0.076	0.541	0.000	1.354	0.000	0.000
combined shift	Particle belief	192	0.573	0.077	0.509	0.000	1.566	0.000	0.000
combined shift	Old VT branch	192	0.495	0.083	0.552	0.000	0.774	0.505	0.000
combined shift	CVTB-MPC	192	0.547	0.078	0.509	0.000	1.206	0.000	0.000

Split	Method	Ep.	Succ.	Err.	Eng.	Viol.	Ent.	Diag.	Fb.
combined shift	CVTB no-probe	192	0.547	0.078	0.509	0.000	1.206	0.000	0.000
combined shift	Oracle	192	0.859	0.050	0.552	0.000	0.000	0.000	0.000
contact dropout	Random	192	0.125	0.143	0.363	0.000	0.000	0.000	0.000
contact dropout	Vision only	192	0.927	0.035	0.459	0.000	0.000	0.000	0.000
contact dropout	Tactile only	192	0.849	0.040	0.458	0.000	0.000	0.000	0.000
contact dropout	Mean fusion	192	0.911	0.035	0.459	0.000	0.000	0.000	0.000
contact dropout	Ensemble uncertainty	192	0.849	0.041	0.458	0.000	1.077	0.000	0.000
contact dropout	Conformal risk	192	0.880	0.043	0.448	0.000	0.000	0.000	0.000
contact dropout	Diagnostic probe	192	0.818	0.043	0.540	0.000	0.000	1.000	0.000
contact dropout	Robust minimax	192	0.875	0.048	0.490	0.000	1.354	0.000	0.000
contact dropout	Particle belief	192	0.891	0.037	0.459	0.000	1.566	0.000	0.000
contact dropout	Old VT branch	192	0.802	0.045	0.487	0.000	0.778	0.359	0.000
contact dropout	CVTB-MPC	192	0.839	0.059	0.424	0.000	1.138	0.000	1.000
contact dropout	CVTB no-probe	192	0.839	0.059	0.424	0.000	1.138	0.000	1.000
contact dropout	Oracle	192	0.948	0.041	0.456	0.000	0.000	0.000	0.000
delayed touch sticky	Random	192	0.047	0.173	0.591	0.000	0.000	0.000	0.000
delayed touch sticky	Vision only	192	0.620	0.084	0.757	0.000	0.000	0.000	0.000
delayed touch sticky	Tactile only	192	0.625	0.089	0.755	0.000	0.000	0.000	0.000
delayed touch sticky	Mean fusion	192	0.635	0.083	0.761	0.000	0.000	0.000	0.000
delayed touch sticky	Ensemble uncertainty	192	0.635	0.088	0.753	0.000	1.077	0.000	0.000
delayed touch sticky	Conformal risk	192	0.630	0.083	0.756	0.000	0.000	0.000	0.000
delayed touch sticky	Diagnostic probe	192	0.609	0.086	0.883	0.000	0.000	1.000	0.000
delayed touch sticky	Robust minimax	192	0.646	0.074	0.777	0.000	1.354	0.000	0.000
delayed touch sticky	Particle belief	192	0.630	0.087	0.755	0.000	1.566	0.000	0.000
delayed touch sticky	Old VT branch	192	0.641	0.087	0.756	0.000	0.840	0.010	0.000
delayed touch sticky	CVTB-MPC	192	0.625	0.085	0.756	0.000	1.198	0.000	0.010
delayed touch sticky	CVTB no-probe	192	0.625	0.085	0.756	0.000	1.198	0.000	0.010
delayed touch sticky	Oracle	192	0.922	0.043	0.888	0.000	0.000	0.000	0.000
high friction	Random	192	0.323	0.097	0.440	0.000	0.000	0.000	0.000
high friction	Vision only	192	1.000	0.021	0.490	0.000	0.000	0.000	0.000
high friction	Tactile only	192	1.000	0.019	0.489	0.000	0.000	0.000	0.000
high friction	Mean fusion	192	1.000	0.020	0.490	0.000	0.000	0.000	0.000
high friction	Ensemble uncertainty	192	1.000	0.019	0.491	0.000	1.077	0.000	0.000
high friction	Conformal risk	192	1.000	0.019	0.492	0.000	0.000	0.000	0.000
high friction	Diagnostic probe	192	1.000	0.018	0.573	0.000	0.000	1.000	0.000
high friction	Robust minimax	192	1.000	0.023	0.524	0.000	1.354	0.000	0.000
high friction	Particle belief	192	1.000	0.019	0.491	0.000	1.566	0.000	0.000
high friction	Old VT branch	192	1.000	0.019	0.488	0.000	0.843	0.000	0.000
high friction	CVTB-MPC	192	1.000	0.019	0.490	0.000	1.166	0.000	0.000
high friction	CVTB no-probe	192	1.000	0.019	0.490	0.000	1.166	0.000	0.000
high friction	Oracle	192	1.000	0.030	0.514	0.000	0.000	0.000	0.000
low friction	Random	192	0.365	0.090	0.337	0.000	0.000	0.000	0.000
low friction	Vision only	192	1.000	0.011	0.370	0.000	0.000	0.000	0.000
low friction	Tactile only	192	1.000	0.013	0.365	0.000	0.000	0.000	0.000
low friction	Mean fusion	192	1.000	0.012	0.367	0.000	0.000	0.000	0.000
low friction	Ensemble uncertainty	192	1.000	0.012	0.366	0.000	1.077	0.000	0.000
low friction	Conformal risk	192	1.000	0.013	0.370	0.000	0.000	0.000	0.000
low friction	Diagnostic probe	192	1.000	0.012	0.421	0.000	0.000	1.000	0.000
low friction	Robust minimax	192	0.974	0.039	0.394	0.000	1.354	0.000	0.000
low friction	Particle belief	192	1.000	0.012	0.365	0.000	1.566	0.000	0.000
low friction	Old VT branch	192	1.000	0.013	0.364	0.000	0.843	0.000	0.000
low friction	CVTB-MPC	192	1.000	0.012	0.368	0.000	1.166	0.000	0.000
low friction	CVTB no-probe	192	1.000	0.012	0.368	0.000	1.166	0.000	0.000
low friction	Oracle	192	1.000	0.029	0.324	0.000	0.000	0.000	0.000
nominal	Random	192	0.333	0.097	0.370	0.000	0.000	0.000	0.000
nominal	Vision only	192	1.000	0.013	0.408	0.000	0.000	0.000	0.000
nominal	Tactile only	192	1.000	0.011	0.405	0.000	0.000	0.000	0.000
nominal	Mean fusion	192	1.000	0.012	0.405	0.000	0.000	0.000	0.000
nominal	Ensemble uncertainty	192	1.000	0.011	0.406	0.000	1.077	0.000	0.000
nominal	Conformal risk	192	1.000	0.011	0.408	0.000	0.000	0.000	0.000
nominal	Diagnostic probe	192	1.000	0.012	0.471	0.000	0.000	1.000	0.000
nominal	Robust minimax	192	1.000	0.029	0.434	0.000	1.354	0.000	0.000
nominal	Particle belief	192	1.000	0.011	0.404	0.000	1.566	0.000	0.000
nominal	Old VT branch	192	1.000	0.011	0.403	0.000	0.844	0.000	0.000
nominal	CVTB-MPC	192	1.000	0.011	0.406	0.000	1.166	0.000	0.000
nominal	CVTB no-probe	192	1.000	0.011	0.406	0.000	1.166	0.000	0.000
nominal	Oracle	192	1.000	0.028	0.391	0.000	0.000	0.000	0.000
sensor conflict	Random	192	0.010	0.202	0.353	0.000	0.000	0.000	0.000
sensor conflict	Vision only	192	0.380	0.143	0.439	0.000	0.000	0.000	0.000
sensor conflict	Tactile only	192	0.286	0.168	0.417	0.000	0.000	0.000	0.000
sensor conflict	Mean fusion	192	0.375	0.143	0.430	0.000	0.000	0.000	0.000
sensor conflict	Ensemble uncertainty	192	0.318	0.155	0.421	0.000	1.077	0.000	0.000
sensor conflict	Conformal risk	192	0.281	0.163	0.393	0.000	0.000	0.000	0.000
sensor conflict	Diagnostic probe	192	0.281	0.170	0.505	0.000	0.000	1.000	0.000
sensor conflict	Robust minimax	192	0.323	0.147	0.452	0.000	1.354	0.000	0.000
sensor conflict	Particle belief	192	0.344	0.149	0.430	0.000	1.566	0.000	0.000
sensor conflict	Old VT branch	192	0.286	0.164	0.486	0.000	0.684	0.818	0.000
sensor conflict	CVTB-MPC	192	0.312	0.151	0.418	0.000	1.253	0.000	0.255
sensor conflict	CVTB no-probe	192	0.312	0.151	0.418	0.000	1.253	0.000	0.255

Split	Method	Ep.	Succ.	Err.	Eng.	Viol.	Ent.	Diag.	Fb.
sensor conflict	Oracle	192	0.578	0.072	0.460	0.000	0.000	0.000	0.000
sticky contact	Random	192	0.271	0.107	0.582	0.000	0.000	0.000	0.000
sticky contact	Vision only	192	1.000	0.037	0.670	0.000	0.000	0.000	0.000
sticky contact	Tactile only	192	0.995	0.035	0.679	0.000	0.000	0.000	0.000
sticky contact	Mean fusion	192	1.000	0.035	0.674	0.000	0.000	0.000	0.000
sticky contact	Ensemble uncertainty	192	1.000	0.035	0.676	0.000	1.077	0.000	0.000
sticky contact	Conformal risk	192	1.000	0.034	0.676	0.000	0.000	0.000	0.000
sticky contact	Diagnostic probe	192	1.000	0.032	0.803	0.000	0.000	1.000	0.000
sticky contact	Robust minimax	192	1.000	0.022	0.721	0.000	1.354	0.000	0.000
sticky contact	Particle belief	192	1.000	0.034	0.679	0.000	1.566	0.000	0.000
sticky contact	Old VT branch	192	0.995	0.035	0.677	0.000	0.840	0.000	0.000
sticky contact	CVTB-MPC	192	1.000	0.035	0.676	0.000	1.168	0.000	0.000
sticky contact	CVTB no-probe	192	1.000	0.035	0.676	0.000	1.168	0.000	0.000
sticky contact	Oracle	192	1.000	0.023	0.760	0.000	0.000	0.000	0.000
tactile noise	Random	192	0.307	0.103	0.367	0.000	0.000	0.000	0.000
tactile noise	Vision only	192	1.000	0.011	0.424	0.000	0.000	0.000	0.000
tactile noise	Tactile only	192	0.859	0.037	0.420	0.000	0.000	0.000	0.000
tactile noise	Mean fusion	192	0.995	0.014	0.424	0.000	0.000	0.000	0.000
tactile noise	Ensemble uncertainty	192	0.964	0.021	0.424	0.000	1.077	0.000	0.000
tactile noise	Conformal risk	192	0.922	0.031	0.406	0.000	0.000	0.000	0.000
tactile noise	Diagnostic probe	192	0.802	0.042	0.503	0.000	0.000	1.000	0.000
tactile noise	Robust minimax	192	1.000	0.039	0.459	0.000	1.354	0.000	0.000
tactile noise	Particle belief	192	0.990	0.016	0.427	0.000	1.566	0.000	0.000
tactile noise	Old VT branch	192	0.839	0.037	0.476	0.000	0.749	0.625	0.000
tactile noise	CVTB-MPC	192	0.974	0.018	0.423	0.000	1.232	0.000	0.094
tactile noise	CVTB no-probe	192	0.974	0.018	0.423	0.000	1.232	0.000	0.094
tactile noise	Oracle	192	0.969	0.039	0.403	0.000	0.000	0.000	0.000
vision bias	Random	192	0.146	0.143	0.358	0.000	0.000	0.000	0.000
vision bias	Vision only	192	0.474	0.088	0.405	0.000	0.000	0.000	0.000
vision bias	Tactile only	192	0.469	0.087	0.387	0.000	0.000	0.000	0.000
vision bias	Mean fusion	192	0.474	0.088	0.396	0.000	0.000	0.000	0.000
vision bias	Ensemble uncertainty	192	0.474	0.088	0.393	0.000	1.077	0.000	0.000
vision bias	Conformal risk	192	0.469	0.092	0.386	0.000	0.000	0.000	0.000
vision bias	Diagnostic probe	192	0.474	0.087	0.470	0.000	0.000	1.000	0.000
vision bias	Robust minimax	192	0.438	0.098	0.426	0.000	1.354	0.000	0.000
vision bias	Particle belief	192	0.469	0.088	0.396	0.000	1.566	0.000	0.000
vision bias	Old VT branch	192	0.464	0.088	0.421	0.000	0.788	0.349	0.000
vision bias	CVTB-MPC	192	0.469	0.088	0.395	0.000	1.176	0.000	0.047
vision bias	CVTB no-probe	192	0.469	0.088	0.395	0.000	1.176	0.000	0.047
vision bias	Oracle	192	0.807	0.053	0.401	0.000	0.000	0.000	0.000

F Full Paired Comparisons

Table 10: Complete paired CVTB-MPC comparison table.

Split	Baseline	Pairs	Δ succ.	Err. impr.	Eng. impr.	Viol. Δ	Diff.
combined shift	Random	192	0.406	0.062	-0.079	0.000	100.0%
combined shift	Vision only	192	-0.062	-0.007	0.007	0.000	56.8%
combined shift	Tactile only	192	0.042	0.005	-0.010	0.000	40.1%
combined shift	Mean fusion	192	-0.026	-0.003	0.002	0.000	0.5%
combined shift	Ensemble uncertainty	192	0.005	0.002	-0.003	0.000	0.0%
combined shift	Conformal risk	192	0.052	0.007	-0.015	0.000	33.3%
combined shift	Diagnostic probe	192	0.031	0.003	0.089	0.000	100.0%
combined shift	Robust minimax	192	-0.036	-0.002	0.032	0.000	77.6%
combined shift	Particle belief	192	-0.026	-0.001	0.001	0.000	0.5%
combined shift	Old VT branch	192	0.052	0.005	0.043	0.000	50.5%
combined shift	CVTB no-probe	192	0.000	0.000	0.000	0.000	0.0%
combined shift	Oracle	192	-0.312	-0.028	0.043	0.000	97.4%
contact dropout	Random	192	0.714	0.084	-0.061	0.000	99.5%
contact dropout	Vision only	192	-0.088	-0.023	0.035	0.000	94.8%
contact dropout	Tactile only	192	-0.010	-0.019	0.034	0.000	98.4%
contact dropout	Mean fusion	192	-0.073	-0.024	0.035	0.000	100.0%
contact dropout	Ensemble uncertainty	192	-0.010	-0.018	0.034	0.000	91.1%
contact dropout	Conformal risk	192	-0.042	-0.016	0.024	0.000	82.3%
contact dropout	Diagnostic probe	192	0.021	-0.016	0.116	0.000	100.0%
contact dropout	Robust minimax	192	-0.036	-0.011	0.066	0.000	100.0%
contact dropout	Particle belief	192	-0.052	-0.022	0.036	0.000	96.4%
contact dropout	Old VT branch	192	0.036	-0.014	0.063	0.000	98.4%
contact dropout	CVTB no-probe	192	0.000	0.000	0.000	0.000	0.0%
contact dropout	Oracle	192	-0.109	-0.018	0.032	0.000	100.0%
delayed touch sticky	Random	192	0.578	0.087	-0.165	0.000	100.0%
delayed touch sticky	Vision only	192	0.005	-0.001	0.001	0.000	1.6%
delayed touch sticky	Tactile only	192	0.000	0.004	-0.001	0.000	1.0%
delayed touch sticky	Mean fusion	192	-0.010	-0.003	0.005	0.000	1.0%
delayed touch sticky	Ensemble uncertainty	192	-0.010	0.003	-0.003	0.000	1.6%
delayed touch sticky	Conformal risk	192	-0.005	-0.002	0.000	0.000	0.0%
delayed touch sticky	Diagnostic probe	192	0.016	0.000	0.128	0.000	100.0%
delayed touch sticky	Robust minimax	192	-0.021	-0.011	0.021	0.000	40.6%
delayed touch sticky	Particle belief	192	-0.005	0.002	-0.001	0.000	1.0%
delayed touch sticky	Old VT branch	192	-0.016	0.002	-0.000	0.000	1.0%
delayed touch sticky	CVTB no-probe	192	0.000	0.000	0.000	0.000	0.0%
delayed touch sticky	Oracle	192	-0.297	-0.043	0.132	0.000	100.0%
high friction	Random	192	0.677	0.077	-0.051	0.000	98.4%
high friction	Vision only	192	0.000	0.002	-0.000	0.000	0.5%
high friction	Tactile only	192	0.000	-0.000	-0.001	0.000	0.0%
high friction	Mean fusion	192	0.000	0.001	-0.000	0.000	0.0%
high friction	Ensemble uncertainty	192	0.000	0.000	0.000	0.000	0.0%
high friction	Conformal risk	192	0.000	-0.000	0.001	0.000	0.0%
high friction	Diagnostic probe	192	0.000	-0.001	0.082	0.000	100.0%
high friction	Robust minimax	192	0.000	0.004	0.033	0.000	58.3%
high friction	Particle belief	192	0.000	-0.000	0.000	0.000	0.0%
high friction	Old VT branch	192	0.000	0.000	-0.002	0.000	0.0%
high friction	CVTB no-probe	192	0.000	0.000	0.000	0.000	0.0%
high friction	Oracle	192	0.000	0.011	0.024	0.000	100.0%
low friction	Random	192	0.635	0.078	-0.032	0.000	98.4%
low friction	Vision only	192	0.000	-0.001	0.002	0.000	0.0%
low friction	Tactile only	192	0.000	0.001	-0.004	0.000	0.0%
low friction	Mean fusion	192	0.000	-0.001	-0.001	0.000	0.0%
low friction	Ensemble uncertainty	192	0.000	-0.000	-0.002	0.000	0.0%
low friction	Conformal risk	192	0.000	0.000	0.001	0.000	0.0%
low friction	Diagnostic probe	192	0.000	-0.000	0.053	0.000	100.0%
low friction	Robust minimax	192	0.026	0.027	0.026	0.000	55.2%
low friction	Particle belief	192	0.000	-0.000	-0.004	0.000	0.0%
low friction	Old VT branch	192	0.000	0.000	-0.004	0.000	0.0%
low friction	CVTB no-probe	192	0.000	0.000	0.000	0.000	0.0%
low friction	Oracle	192	0.000	0.016	-0.044	0.000	100.0%
nominal	Random	192	0.667	0.085	-0.036	0.000	100.0%
nominal	Vision only	192	0.000	0.001	0.001	0.000	0.0%
nominal	Tactile only	192	0.000	0.000	-0.002	0.000	0.0%
nominal	Mean fusion	192	0.000	0.001	-0.001	0.000	0.0%
nominal	Ensemble uncertainty	192	0.000	0.000	-0.001	0.000	0.0%
nominal	Conformal risk	192	0.000	-0.000	0.002	0.000	0.0%
nominal	Diagnostic probe	192	0.000	0.001	0.065	0.000	100.0%
nominal	Robust minimax	192	0.000	0.018	0.028	0.000	54.7%
nominal	Particle belief	192	0.000	0.000	-0.003	0.000	0.0%
nominal	Old VT branch	192	0.000	0.000	-0.003	0.000	0.0%
nominal	CVTB no-probe	192	0.000	0.000	0.000	0.000	0.0%
nominal	Oracle	192	0.000	0.016	-0.015	0.000	100.0%
sensor conflict	Random	192	0.302	0.050	-0.065	0.000	100.0%
sensor conflict	Vision only	192	-0.068	-0.009	0.020	0.000	87.0%

Split	Baseline	Pairs	Δ succ.	Err. impr.	Eng. impr.	Viol. Δ	Diff.
sensor conflict	Tactile only	192	0.026	0.017	-0.002	0.000	85.4%
sensor conflict	Mean fusion	192	-0.062	-0.008	0.012	0.000	39.1%
sensor conflict	Ensemble uncertainty	192	-0.005	0.004	0.003	0.000	25.5%
sensor conflict	Conformal risk	192	0.031	0.012	-0.026	0.000	63.0%
sensor conflict	Diagnostic probe	192	0.031	0.018	0.087	0.000	100.0%
sensor conflict	Robust minimax	192	-0.010	-0.004	0.034	0.000	71.9%
sensor conflict	Particle belief	192	-0.031	-0.002	0.011	0.000	22.9%
sensor conflict	Old VT branch	192	0.026	0.013	0.068	0.000	85.9%
sensor conflict	CVTB no-probe	192	0.000	0.000	0.000	0.000	0.0%
sensor conflict	Oracle	192	-0.266	-0.079	0.042	0.000	99.0%
sticky contact	Random	192	0.729	0.072	-0.094	0.000	100.0%
sticky contact	Vision only	192	0.000	0.002	-0.006	0.000	2.1%
sticky contact	Tactile only	192	0.005	0.000	0.003	0.000	0.0%
sticky contact	Mean fusion	192	0.000	0.001	-0.002	0.000	0.0%
sticky contact	Ensemble uncertainty	192	0.000	0.000	-0.000	0.000	0.0%
sticky contact	Conformal risk	192	0.000	-0.000	0.000	0.000	0.0%
sticky contact	Diagnostic probe	192	0.000	-0.002	0.127	0.000	100.0%
sticky contact	Robust minimax	192	0.000	-0.013	0.045	0.000	62.0%
sticky contact	Particle belief	192	0.000	-0.000	0.003	0.000	0.0%
sticky contact	Old VT branch	192	0.005	0.001	0.001	0.000	0.0%
sticky contact	CVTB no-probe	192	0.000	0.000	0.000	0.000	0.0%
sticky contact	Oracle	192	0.000	-0.011	0.084	0.000	100.0%
tactile noise	Random	192	0.667	0.085	-0.056	0.000	99.5%
tactile noise	Vision only	192	-0.026	-0.007	0.001	0.000	64.6%
tactile noise	Tactile only	192	0.115	0.018	-0.003	0.000	60.4%
tactile noise	Mean fusion	192	-0.021	-0.004	0.001	0.000	12.0%
tactile noise	Ensemble uncertainty	192	0.010	0.002	0.001	0.000	8.8%
tactile noise	Conformal risk	192	0.052	0.013	-0.017	0.000	46.4%
tactile noise	Diagnostic probe	192	0.172	0.023	0.080	0.000	100.0%
tactile noise	Robust minimax	192	-0.026	0.021	0.036	0.000	81.8%
tactile noise	Particle belief	192	-0.016	-0.002	0.004	0.000	10.9%
tactile noise	Old VT branch	192	0.135	0.019	0.053	0.000	63.0%
tactile noise	CVTB no-probe	192	0.000	0.000	0.000	0.000	0.0%
tactile noise	Oracle	192	0.005	0.021	-0.020	0.000	100.0%
vision bias	Random	192	0.323	0.055	-0.037	0.000	99.5%
vision bias	Vision only	192	-0.005	-0.000	0.010	0.000	43.2%
vision bias	Tactile only	192	0.000	-0.000	-0.009	0.000	27.1%
vision bias	Mean fusion	192	-0.005	-0.000	0.000	0.000	3.6%
vision bias	Ensemble uncertainty	192	-0.005	-0.000	-0.002	0.000	3.6%
vision bias	Conformal risk	192	0.000	0.004	-0.009	0.000	25.5%
vision bias	Diagnostic probe	192	-0.005	-0.001	0.075	0.000	100.0%
vision bias	Robust minimax	192	0.031	0.010	0.030	0.000	72.4%
vision bias	Particle belief	192	0.000	0.000	0.000	0.000	4.7%
vision bias	Old VT branch	192	0.005	0.000	0.025	0.000	39.1%
vision bias	CVTB no-probe	192	0.000	0.000	0.000	0.000	0.0%
vision bias	Oracle	192	-0.339	-0.035	0.006	0.000	96.4%

G SEED-LEVEL ROBUSTNESS

Aggregate means can hide seed fragility. Table 11 therefore reports seed-level summaries for selected strong baselines, CVTB-MPC, its no-probe variant, the old branch planner, and the oracle. This table is intentionally kept in the paper rather than only in CSV form. A hostile reviewer should be able to inspect whether a conclusion is carried by one lucky seed or persists across the frozen seed grid.

Table 11: Seed-level robustness table for selected strong baselines, CVTB-MPC, its no-probe variant, and the oracle.

Split	Method	Seed	Ep.	Succ.	Err.	Eng.	Fb.
combined shift	Mean fusion	0	24	0.583	0.073	0.512	0.000
combined shift	Ensemble uncertainty	0	24	0.458	0.083	0.509	0.000
combined shift	Robust minimax	0	24	0.583	0.076	0.546	0.000
combined shift	Old VT branch	0	24	0.333	0.096	0.536	0.000
combined shift	CVTB-MPC	0	24	0.500	0.079	0.512	0.000
combined shift	CVTB no-probe	0	24	0.500	0.079	0.512	0.000
combined shift	Oracle	0	24	0.792	0.054	0.555	0.000
combined shift	Mean fusion	1	24	0.542	0.067	0.529	0.000
combined shift	Ensemble uncertainty	1	24	0.625	0.066	0.529	0.000
combined shift	Robust minimax	1	24	0.583	0.065	0.563	0.000
combined shift	Old VT branch	1	24	0.500	0.072	0.579	0.000
combined shift	CVTB-MPC	1	24	0.542	0.065	0.532	0.000
combined shift	CVTB no-probe	1	24	0.542	0.065	0.532	0.000
combined shift	Oracle	1	24	0.875	0.050	0.565	0.000
combined shift	Mean fusion	2	24	0.542	0.077	0.507	0.000
combined shift	Ensemble uncertainty	2	24	0.458	0.088	0.501	0.000
combined shift	Robust minimax	2	24	0.500	0.084	0.539	0.000
combined shift	Old VT branch	2	24	0.417	0.086	0.544	0.000
combined shift	CVTB-MPC	2	24	0.500	0.086	0.502	0.000
combined shift	CVTB no-probe	2	24	0.500	0.086	0.502	0.000
combined shift	Oracle	2	24	0.792	0.055	0.554	0.000
combined shift	Mean fusion	3	24	0.583	0.070	0.516	0.000
combined shift	Ensemble uncertainty	3	24	0.542	0.074	0.511	0.000
combined shift	Robust minimax	3	24	0.667	0.065	0.545	0.000
combined shift	Old VT branch	3	24	0.542	0.074	0.581	0.000
combined shift	CVTB-MPC	3	24	0.583	0.073	0.510	0.000
combined shift	CVTB no-probe	3	24	0.583	0.073	0.510	0.000
combined shift	Oracle	3	24	0.875	0.050	0.546	0.000
combined shift	Mean fusion	4	24	0.458	0.100	0.484	0.000
combined shift	Ensemble uncertainty	4	24	0.458	0.100	0.480	0.000
combined shift	Robust minimax	4	24	0.500	0.100	0.514	0.000
combined shift	Old VT branch	4	24	0.542	0.097	0.538	0.000
combined shift	CVTB-MPC	4	24	0.500	0.100	0.480	0.000
combined shift	CVTB no-probe	4	24	0.500	0.100	0.480	0.000
combined shift	Oracle	4	24	0.917	0.049	0.550	0.000
combined shift	Mean fusion	5	24	0.667	0.074	0.521	0.000
combined shift	Ensemble uncertainty	5	24	0.625	0.078	0.516	0.000
combined shift	Robust minimax	5	24	0.625	0.077	0.543	0.000
combined shift	Old VT branch	5	24	0.625	0.076	0.545	0.000
combined shift	CVTB-MPC	5	24	0.583	0.075	0.519	0.000
combined shift	CVTB no-probe	5	24	0.583	0.075	0.519	0.000
combined shift	Oracle	5	24	0.875	0.050	0.541	0.000
combined shift	Mean fusion	6	24	0.667	0.072	0.509	0.000
combined shift	Ensemble uncertainty	6	24	0.625	0.080	0.497	0.000
combined shift	Robust minimax	6	24	0.667	0.071	0.534	0.000
combined shift	Old VT branch	6	24	0.458	0.086	0.546	0.000
combined shift	CVTB-MPC	6	24	0.625	0.076	0.505	0.000
combined shift	CVTB no-probe	6	24	0.625	0.076	0.505	0.000
combined shift	Oracle	6	24	0.792	0.049	0.557	0.000
combined shift	Mean fusion	7	24	0.542	0.067	0.508	0.000
combined shift	Ensemble uncertainty	7	24	0.542	0.070	0.504	0.000
combined shift	Robust minimax	7	24	0.542	0.066	0.544	0.000
combined shift	Old VT branch	7	24	0.542	0.075	0.545	0.000
combined shift	CVTB-MPC	7	24	0.542	0.069	0.508	0.000
combined shift	CVTB no-probe	7	24	0.542	0.069	0.508	0.000
combined shift	Oracle	7	24	0.958	0.043	0.550	0.000
contact dropout	Mean fusion	0	24	0.833	0.041	0.457	0.000
contact dropout	Ensemble uncertainty	0	24	0.750	0.052	0.457	0.000
contact dropout	Robust minimax	0	24	0.833	0.055	0.491	0.000
contact dropout	Old VT branch	0	24	0.708	0.053	0.484	0.000
contact dropout	CVTB-MPC	0	24	0.792	0.061	0.427	1.000
contact dropout	CVTB no-probe	0	24	0.792	0.061	0.427	1.000
contact dropout	Oracle	0	24	0.917	0.041	0.464	0.000
contact dropout	Mean fusion	1	24	0.833	0.040	0.451	0.000
contact dropout	Ensemble uncertainty	1	24	0.875	0.042	0.449	0.000

Split	Method	Seed	Ep.	Succ.	Err.	Eng.	Fb.
contact dropout	Robust minimax	1	24	0.833	0.050	0.472	0.000
contact dropout	Old VT branch	1	24	0.792	0.051	0.476	0.000
contact dropout	CVTB-MPC	1	24	0.792	0.061	0.410	1.000
contact dropout	CVTB no-probe	1	24	0.792	0.061	0.410	1.000
contact dropout	Oracle	1	24	0.958	0.047	0.439	0.000
contact dropout	Mean fusion	2	24	0.958	0.035	0.454	0.000
contact dropout	Ensemble uncertainty	2	24	0.833	0.038	0.456	0.000
contact dropout	Robust minimax	2	24	0.917	0.048	0.487	0.000
contact dropout	Old VT branch	2	24	0.833	0.040	0.486	0.000
contact dropout	CVTB-MPC	2	24	0.875	0.059	0.429	1.000
contact dropout	CVTB no-probe	2	24	0.875	0.059	0.429	1.000
contact dropout	Oracle	2	24	0.917	0.039	0.457	0.000
contact dropout	Mean fusion	3	24	1.000	0.027	0.462	0.000
contact dropout	Ensemble uncertainty	3	24	0.917	0.033	0.457	0.000
contact dropout	Robust minimax	3	24	0.958	0.040	0.496	0.000
contact dropout	Old VT branch	3	24	0.875	0.038	0.481	0.000
contact dropout	CVTB-MPC	3	24	1.000	0.050	0.431	1.000
contact dropout	CVTB no-probe	3	24	1.000	0.050	0.431	1.000
contact dropout	Oracle	3	24	0.958	0.040	0.455	0.000
contact dropout	Mean fusion	4	24	0.958	0.034	0.454	0.000
contact dropout	Ensemble uncertainty	4	24	0.917	0.037	0.451	0.000
contact dropout	Robust minimax	4	24	0.833	0.049	0.491	0.000
contact dropout	Old VT branch	4	24	0.833	0.039	0.484	0.000
contact dropout	CVTB-MPC	4	24	0.833	0.058	0.413	1.000
contact dropout	CVTB no-probe	4	24	0.833	0.058	0.413	1.000
contact dropout	Oracle	4	24	1.000	0.037	0.449	0.000
contact dropout	Mean fusion	5	24	0.875	0.036	0.472	0.000
contact dropout	Ensemble uncertainty	5	24	0.792	0.040	0.470	0.000
contact dropout	Robust minimax	5	24	0.875	0.048	0.502	0.000
contact dropout	Old VT branch	5	24	0.833	0.044	0.497	0.000
contact dropout	CVTB-MPC	5	24	0.875	0.059	0.433	1.000
contact dropout	CVTB no-probe	5	24	0.875	0.059	0.433	1.000
contact dropout	Oracle	5	24	0.958	0.044	0.465	0.000
contact dropout	Mean fusion	6	24	0.958	0.029	0.471	0.000
contact dropout	Ensemble uncertainty	6	24	0.875	0.036	0.473	0.000
contact dropout	Robust minimax	6	24	0.875	0.041	0.490	0.000
contact dropout	Old VT branch	6	24	0.833	0.038	0.491	0.000
contact dropout	CVTB-MPC	6	24	0.833	0.057	0.432	1.000
contact dropout	CVTB no-probe	6	24	0.833	0.057	0.432	1.000
contact dropout	Oracle	6	24	0.917	0.039	0.473	0.000
contact dropout	Mean fusion	7	24	0.875	0.041	0.448	0.000
contact dropout	Ensemble uncertainty	7	24	0.833	0.048	0.447	0.000
contact dropout	Robust minimax	7	24	0.875	0.056	0.488	0.000
contact dropout	Old VT branch	7	24	0.708	0.054	0.496	0.000
contact dropout	CVTB-MPC	7	24	0.708	0.067	0.415	1.000
contact dropout	CVTB no-probe	7	24	0.708	0.067	0.415	1.000
contact dropout	Oracle	7	24	0.958	0.039	0.443	0.000
delayed touch sticky	Mean fusion	0	24	0.500	0.096	0.738	0.000
delayed touch sticky	Ensemble uncertainty	0	24	0.500	0.106	0.743	0.000
delayed touch sticky	Robust minimax	0	24	0.500	0.087	0.769	0.000
delayed touch sticky	Old VT branch	0	24	0.500	0.103	0.739	0.000
delayed touch sticky	CVTB-MPC	0	24	0.500	0.106	0.738	0.000
delayed touch sticky	CVTB no-probe	0	24	0.500	0.106	0.738	0.000
delayed touch sticky	Oracle	0	24	0.958	0.041	0.879	0.000
delayed touch sticky	Mean fusion	1	24	0.542	0.097	0.758	0.000
delayed touch sticky	Ensemble uncertainty	1	24	0.500	0.107	0.743	0.000
delayed touch sticky	Robust minimax	1	24	0.500	0.097	0.763	0.000
delayed touch sticky	Old VT branch	1	24	0.458	0.111	0.764	0.000
delayed touch sticky	CVTB-MPC	1	24	0.458	0.107	0.746	0.042
delayed touch sticky	CVTB no-probe	1	24	0.458	0.107	0.746	0.042
delayed touch sticky	Oracle	1	24	0.875	0.052	0.883	0.000
delayed touch sticky	Mean fusion	2	24	0.667	0.072	0.790	0.000
delayed touch sticky	Ensemble uncertainty	2	24	0.625	0.077	0.773	0.000
delayed touch sticky	Robust minimax	2	24	0.792	0.056	0.819	0.000
delayed touch sticky	Old VT branch	2	24	0.625	0.080	0.776	0.000
delayed touch sticky	CVTB-MPC	2	24	0.667	0.072	0.783	0.000
delayed touch sticky	CVTB no-probe	2	24	0.667	0.072	0.783	0.000
delayed touch sticky	Oracle	2	24	0.875	0.042	0.905	0.000
delayed touch sticky	Mean fusion	3	24	0.792	0.069	0.774	0.000
delayed touch sticky	Ensemble uncertainty	3	24	0.750	0.078	0.781	0.000
delayed touch sticky	Robust minimax	3	24	0.792	0.060	0.794	0.000
delayed touch sticky	Old VT branch	3	24	0.833	0.067	0.778	0.000
delayed touch sticky	CVTB-MPC	3	24	0.750	0.070	0.774	0.000
delayed touch sticky	CVTB no-probe	3	24	0.750	0.070	0.774	0.000
delayed touch sticky	Oracle	3	24	0.958	0.042	0.910	0.000
delayed touch sticky	Mean fusion	4	24	0.625	0.075	0.762	0.000
delayed touch sticky	Ensemble uncertainty	4	24	0.708	0.073	0.760	0.000
delayed touch sticky	Robust minimax	4	24	0.625	0.066	0.784	0.000
delayed touch sticky	Old VT branch	4	24	0.667	0.073	0.765	0.000

Split	Method	Seed	Ep.	Succ.	Err.	Eng.	Fb.
delayed touch sticky	CVTB-MPC	4	24	0.625	0.074	0.761	0.000
delayed touch sticky	CVTB no-probe	4	24	0.625	0.074	0.761	0.000
delayed touch sticky	Oracle	4	24	0.958	0.035	0.893	0.000
delayed touch sticky	Mean fusion	5	24	0.458	0.099	0.718	0.000
delayed touch sticky	Ensemble uncertainty	5	24	0.458	0.103	0.709	0.000
delayed touch sticky	Robust minimax	5	24	0.458	0.096	0.735	0.000
delayed touch sticky	Old VT branch	5	24	0.542	0.105	0.706	0.000
delayed touch sticky	CVTB-MPC	5	24	0.458	0.101	0.710	0.042
delayed touch sticky	CVTB no-probe	5	24	0.458	0.101	0.710	0.042
delayed touch sticky	Oracle	5	24	0.875	0.052	0.857	0.000
delayed touch sticky	Mean fusion	6	24	0.583	0.091	0.754	0.000
delayed touch sticky	Ensemble uncertainty	6	24	0.625	0.098	0.735	0.000
delayed touch sticky	Robust minimax	6	24	0.583	0.082	0.761	0.000
delayed touch sticky	Old VT branch	6	24	0.625	0.090	0.739	0.000
delayed touch sticky	CVTB-MPC	6	24	0.625	0.093	0.741	0.000
delayed touch sticky	CVTB no-probe	6	24	0.625	0.093	0.741	0.000
delayed touch sticky	Oracle	6	24	0.917	0.045	0.891	0.000
delayed touch sticky	Mean fusion	7	24	0.917	0.061	0.790	0.000
delayed touch sticky	Ensemble uncertainty	7	24	0.917	0.063	0.782	0.000
delayed touch sticky	Robust minimax	7	24	0.917	0.054	0.795	0.000
delayed touch sticky	Old VT branch	7	24	0.875	0.065	0.778	0.000
delayed touch sticky	CVTB-MPC	7	24	0.917	0.060	0.795	0.000
delayed touch sticky	CVTB no-probe	7	24	0.917	0.060	0.795	0.000
delayed touch sticky	Oracle	7	24	0.958	0.033	0.887	0.000
high friction	Mean fusion	0	24	1.000	0.019	0.497	0.000
high friction	Ensemble uncertainty	0	24	1.000	0.019	0.496	0.000
high friction	Robust minimax	0	24	1.000	0.025	0.530	0.000
high friction	Old VT branch	0	24	1.000	0.019	0.492	0.000
high friction	CVTB-MPC	0	24	1.000	0.018	0.495	0.000
high friction	CVTB no-probe	0	24	1.000	0.018	0.495	0.000
high friction	Oracle	0	24	1.000	0.030	0.517	0.000
high friction	Mean fusion	1	24	1.000	0.020	0.487	0.000
high friction	Ensemble uncertainty	1	24	1.000	0.020	0.489	0.000
high friction	Robust minimax	1	24	1.000	0.023	0.520	0.000
high friction	Old VT branch	1	24	1.000	0.020	0.485	0.000
high friction	CVTB-MPC	1	24	1.000	0.019	0.486	0.000
high friction	CVTB no-probe	1	24	1.000	0.019	0.486	0.000
high friction	Oracle	1	24	1.000	0.029	0.507	0.000
high friction	Mean fusion	2	24	1.000	0.020	0.496	0.000
high friction	Ensemble uncertainty	2	24	1.000	0.019	0.498	0.000
high friction	Robust minimax	2	24	1.000	0.019	0.524	0.000
high friction	Old VT branch	2	24	1.000	0.019	0.495	0.000
high friction	CVTB-MPC	2	24	1.000	0.019	0.502	0.000
high friction	CVTB no-probe	2	24	1.000	0.019	0.502	0.000
high friction	Oracle	2	24	1.000	0.031	0.524	0.000
high friction	Mean fusion	3	24	1.000	0.020	0.492	0.000
high friction	Ensemble uncertainty	3	24	1.000	0.020	0.492	0.000
high friction	Robust minimax	3	24	1.000	0.020	0.523	0.000
high friction	Old VT branch	3	24	1.000	0.020	0.490	0.000
high friction	CVTB-MPC	3	24	1.000	0.019	0.492	0.000
high friction	CVTB no-probe	3	24	1.000	0.019	0.492	0.000
high friction	Oracle	3	24	1.000	0.030	0.516	0.000
high friction	Mean fusion	4	24	1.000	0.020	0.491	0.000
high friction	Ensemble uncertainty	4	24	1.000	0.020	0.494	0.000
high friction	Robust minimax	4	24	1.000	0.022	0.522	0.000
high friction	Old VT branch	4	24	1.000	0.020	0.489	0.000
high friction	CVTB-MPC	4	24	1.000	0.019	0.492	0.000
high friction	CVTB no-probe	4	24	1.000	0.019	0.492	0.000
high friction	Oracle	4	24	1.000	0.031	0.521	0.000
high friction	Mean fusion	5	24	1.000	0.020	0.482	0.000
high friction	Ensemble uncertainty	5	24	1.000	0.019	0.479	0.000
high friction	Robust minimax	5	24	1.000	0.023	0.517	0.000
high friction	Old VT branch	5	24	1.000	0.019	0.480	0.000
high friction	CVTB-MPC	5	24	1.000	0.019	0.482	0.000
high friction	CVTB no-probe	5	24	1.000	0.019	0.482	0.000
high friction	Oracle	5	24	1.000	0.029	0.504	0.000
high friction	Mean fusion	6	24	1.000	0.021	0.493	0.000
high friction	Ensemble uncertainty	6	24	1.000	0.020	0.491	0.000
high friction	Robust minimax	6	24	1.000	0.028	0.531	0.000
high friction	Old VT branch	6	24	1.000	0.020	0.493	0.000
high friction	CVTB-MPC	6	24	1.000	0.020	0.492	0.000
high friction	CVTB no-probe	6	24	1.000	0.020	0.492	0.000
high friction	Oracle	6	24	1.000	0.031	0.515	0.000
high friction	Mean fusion	7	24	1.000	0.020	0.484	0.000
high friction	Ensemble uncertainty	7	24	1.000	0.019	0.486	0.000
high friction	Robust minimax	7	24	1.000	0.025	0.521	0.000
high friction	Old VT branch	7	24	1.000	0.019	0.482	0.000
high friction	CVTB-MPC	7	24	1.000	0.019	0.482	0.000
high friction	CVTB no-probe	7	24	1.000	0.019	0.482	0.000

Split	Method	Seed	Ep.	Succ.	Err.	Eng.	Fb.
high friction	Oracle	7	24	1.000	0.030	0.509	0.000
low friction	Mean fusion	0	24	1.000	0.013	0.366	0.000
low friction	Ensemble uncertainty	0	24	1.000	0.013	0.365	0.000
low friction	Robust minimax	0	24	1.000	0.040	0.394	0.000
low friction	Old VT branch	0	24	1.000	0.014	0.362	0.000
low friction	CVTB-MPC	0	24	1.000	0.013	0.368	0.000
low friction	CVTB no-probe	0	24	1.000	0.013	0.368	0.000
low friction	Oracle	0	24	1.000	0.030	0.323	0.000
low friction	Mean fusion	1	24	1.000	0.012	0.363	0.000
low friction	Ensemble uncertainty	1	24	1.000	0.011	0.362	0.000
low friction	Robust minimax	1	24	1.000	0.040	0.389	0.000
low friction	Old VT branch	1	24	1.000	0.012	0.360	0.000
low friction	CVTB-MPC	1	24	1.000	0.011	0.364	0.000
low friction	CVTB no-probe	1	24	1.000	0.011	0.364	0.000
low friction	Oracle	1	24	1.000	0.028	0.319	0.000
low friction	Mean fusion	2	24	1.000	0.011	0.362	0.000
low friction	Ensemble uncertainty	2	24	1.000	0.012	0.361	0.000
low friction	Robust minimax	2	24	0.875	0.038	0.387	0.000
low friction	Old VT branch	2	24	1.000	0.012	0.359	0.000
low friction	CVTB-MPC	2	24	1.000	0.011	0.363	0.000
low friction	CVTB no-probe	2	24	1.000	0.011	0.363	0.000
low friction	Oracle	2	24	1.000	0.027	0.319	0.000
low friction	Mean fusion	3	24	1.000	0.013	0.362	0.000
low friction	Ensemble uncertainty	3	24	1.000	0.013	0.361	0.000
low friction	Robust minimax	3	24	0.958	0.038	0.389	0.000
low friction	Old VT branch	3	24	1.000	0.013	0.357	0.000
low friction	CVTB-MPC	3	24	1.000	0.013	0.363	0.000
low friction	CVTB no-probe	3	24	1.000	0.013	0.363	0.000
low friction	Oracle	3	24	1.000	0.028	0.320	0.000
low friction	Mean fusion	4	24	1.000	0.011	0.378	0.000
low friction	Ensemble uncertainty	4	24	1.000	0.011	0.377	0.000
low friction	Robust minimax	4	24	0.958	0.034	0.401	0.000
low friction	Old VT branch	4	24	1.000	0.012	0.376	0.000
low friction	CVTB-MPC	4	24	1.000	0.011	0.378	0.000
low friction	CVTB no-probe	4	24	1.000	0.011	0.378	0.000
low friction	Oracle	4	24	1.000	0.030	0.332	0.000
low friction	Mean fusion	5	24	1.000	0.013	0.370	0.000
low friction	Ensemble uncertainty	5	24	1.000	0.013	0.369	0.000
low friction	Robust minimax	5	24	1.000	0.041	0.399	0.000
low friction	Old VT branch	5	24	1.000	0.014	0.367	0.000
low friction	CVTB-MPC	5	24	1.000	0.014	0.371	0.000
low friction	CVTB no-probe	5	24	1.000	0.014	0.371	0.000
low friction	Oracle	5	24	1.000	0.030	0.330	0.000
low friction	Mean fusion	6	24	1.000	0.011	0.356	0.000
low friction	Ensemble uncertainty	6	24	1.000	0.012	0.355	0.000
low friction	Robust minimax	6	24	1.000	0.041	0.382	0.000
low friction	Old VT branch	6	24	1.000	0.012	0.354	0.000
low friction	CVTB-MPC	6	24	1.000	0.012	0.357	0.000
low friction	CVTB no-probe	6	24	1.000	0.012	0.357	0.000
low friction	Oracle	6	24	1.000	0.026	0.309	0.000
low friction	Mean fusion	7	24	1.000	0.012	0.381	0.000
low friction	Ensemble uncertainty	7	24	1.000	0.013	0.380	0.000
low friction	Robust minimax	7	24	1.000	0.042	0.411	0.000
low friction	Old VT branch	7	24	1.000	0.013	0.377	0.000
low friction	CVTB-MPC	7	24	1.000	0.013	0.382	0.000
low friction	CVTB no-probe	7	24	1.000	0.013	0.382	0.000
low friction	Oracle	7	24	1.000	0.031	0.339	0.000
nominal	Mean fusion	0	24	1.000	0.013	0.402	0.000
nominal	Ensemble uncertainty	0	24	1.000	0.013	0.399	0.000
nominal	Robust minimax	0	24	1.000	0.035	0.435	0.000
nominal	Old VT branch	0	24	1.000	0.012	0.395	0.000
nominal	CVTB-MPC	0	24	1.000	0.012	0.404	0.000
nominal	CVTB no-probe	0	24	1.000	0.012	0.404	0.000
nominal	Oracle	0	24	1.000	0.027	0.389	0.000
nominal	Mean fusion	1	24	1.000	0.011	0.403	0.000
nominal	Ensemble uncertainty	1	24	1.000	0.011	0.404	0.000
nominal	Robust minimax	1	24	1.000	0.028	0.433	0.000
nominal	Old VT branch	1	24	1.000	0.011	0.401	0.000
nominal	CVTB-MPC	1	24	1.000	0.010	0.401	0.000
nominal	CVTB no-probe	1	24	1.000	0.010	0.401	0.000
nominal	Oracle	1	24	1.000	0.027	0.391	0.000
nominal	Mean fusion	2	24	1.000	0.012	0.401	0.000
nominal	Ensemble uncertainty	2	24	1.000	0.012	0.400	0.000
nominal	Robust minimax	2	24	1.000	0.030	0.424	0.000
nominal	Old VT branch	2	24	1.000	0.012	0.398	0.000
nominal	CVTB-MPC	2	24	1.000	0.012	0.401	0.000
nominal	CVTB no-probe	2	24	1.000	0.012	0.401	0.000
nominal	Oracle	2	24	1.000	0.026	0.382	0.000
nominal	Mean fusion	3	24	1.000	0.010	0.416	0.000

Split	Method	Seed	Ep.	Succ.	Err.	Eng.	Fb.
nominal	Ensemble uncertainty	3	24	1.000	0.010	0.414	0.000
nominal	Robust minimax	3	24	1.000	0.029	0.441	0.000
nominal	Old VT branch	3	24	1.000	0.011	0.411	0.000
nominal	CVTB-MPC	3	24	1.000	0.010	0.415	0.000
nominal	CVTB no-probe	3	24	1.000	0.010	0.415	0.000
nominal	Oracle	3	24	1.000	0.029	0.400	0.000
nominal	Mean fusion	4	24	1.000	0.011	0.401	0.000
nominal	Ensemble uncertainty	4	24	1.000	0.011	0.402	0.000
nominal	Robust minimax	4	24	1.000	0.029	0.426	0.000
nominal	Old VT branch	4	24	1.000	0.010	0.399	0.000
nominal	CVTB-MPC	4	24	1.000	0.011	0.404	0.000
nominal	CVTB no-probe	4	24	1.000	0.011	0.404	0.000
nominal	Oracle	4	24	1.000	0.027	0.388	0.000
nominal	Mean fusion	5	24	1.000	0.013	0.403	0.000
nominal	Ensemble uncertainty	5	24	1.000	0.012	0.402	0.000
nominal	Robust minimax	5	24	1.000	0.029	0.435	0.000
nominal	Old VT branch	5	24	1.000	0.012	0.402	0.000
nominal	CVTB-MPC	5	24	1.000	0.013	0.403	0.000
nominal	CVTB no-probe	5	24	1.000	0.013	0.403	0.000
nominal	Oracle	5	24	1.000	0.028	0.392	0.000
nominal	Mean fusion	6	24	1.000	0.012	0.404	0.000
nominal	Ensemble uncertainty	6	24	1.000	0.010	0.410	0.000
nominal	Robust minimax	6	24	1.000	0.024	0.432	0.000
nominal	Old VT branch	6	24	1.000	0.011	0.406	0.000
nominal	CVTB-MPC	6	24	1.000	0.011	0.408	0.000
nominal	CVTB no-probe	6	24	1.000	0.011	0.408	0.000
nominal	Oracle	6	24	1.000	0.028	0.394	0.000
nominal	Mean fusion	7	24	1.000	0.012	0.414	0.000
nominal	Ensemble uncertainty	7	24	1.000	0.013	0.416	0.000
nominal	Robust minimax	7	24	1.000	0.029	0.445	0.000
nominal	Old VT branch	7	24	1.000	0.012	0.414	0.000
nominal	CVTB-MPC	7	24	1.000	0.011	0.415	0.000
nominal	CVTB no-probe	7	24	1.000	0.011	0.415	0.000
nominal	Oracle	7	24	1.000	0.029	0.396	0.000
sensor conflict	Mean fusion	0	24	0.375	0.137	0.423	0.000
sensor conflict	Ensemble uncertainty	0	24	0.458	0.134	0.414	0.000
sensor conflict	Robust minimax	0	24	0.375	0.142	0.439	0.000
sensor conflict	Old VT branch	0	24	0.375	0.160	0.493	0.000
sensor conflict	CVTB-MPC	0	24	0.375	0.134	0.415	0.250
sensor conflict	CVTB no-probe	0	24	0.375	0.134	0.415	0.250
sensor conflict	Oracle	0	24	0.542	0.074	0.454	0.000
sensor conflict	Mean fusion	1	24	0.542	0.126	0.462	0.000
sensor conflict	Ensemble uncertainty	1	24	0.417	0.144	0.442	0.000
sensor conflict	Robust minimax	1	24	0.417	0.123	0.481	0.000
sensor conflict	Old VT branch	1	24	0.292	0.164	0.495	0.000
sensor conflict	CVTB-MPC	1	24	0.458	0.137	0.445	0.208
sensor conflict	CVTB no-probe	1	24	0.458	0.137	0.445	0.208
sensor conflict	Oracle	1	24	0.542	0.081	0.474	0.000
sensor conflict	Mean fusion	2	24	0.458	0.109	0.420	0.000
sensor conflict	Ensemble uncertainty	2	24	0.375	0.129	0.414	0.000
sensor conflict	Robust minimax	2	24	0.333	0.123	0.447	0.000
sensor conflict	Old VT branch	2	24	0.417	0.129	0.501	0.000
sensor conflict	CVTB-MPC	2	24	0.375	0.124	0.403	0.250
sensor conflict	CVTB no-probe	2	24	0.375	0.124	0.403	0.250
sensor conflict	Oracle	2	24	0.667	0.070	0.450	0.000
sensor conflict	Mean fusion	3	24	0.375	0.129	0.430	0.000
sensor conflict	Ensemble uncertainty	3	24	0.250	0.157	0.423	0.000
sensor conflict	Robust minimax	3	24	0.375	0.132	0.467	0.000
sensor conflict	Old VT branch	3	24	0.208	0.165	0.485	0.000
sensor conflict	CVTB-MPC	3	24	0.333	0.139	0.417	0.250
sensor conflict	CVTB no-probe	3	24	0.333	0.139	0.417	0.250
sensor conflict	Oracle	3	24	0.625	0.064	0.441	0.000
sensor conflict	Mean fusion	4	24	0.417	0.143	0.419	0.000
sensor conflict	Ensemble uncertainty	4	24	0.333	0.153	0.410	0.000
sensor conflict	Robust minimax	4	24	0.417	0.145	0.438	0.000
sensor conflict	Old VT branch	4	24	0.167	0.177	0.464	0.000
sensor conflict	CVTB-MPC	4	24	0.375	0.150	0.398	0.208
sensor conflict	CVTB no-probe	4	24	0.375	0.150	0.398	0.208
sensor conflict	Oracle	4	24	0.500	0.078	0.441	0.000
sensor conflict	Mean fusion	5	24	0.333	0.163	0.418	0.000
sensor conflict	Ensemble uncertainty	5	24	0.333	0.167	0.411	0.000
sensor conflict	Robust minimax	5	24	0.292	0.165	0.446	0.000
sensor conflict	Old VT branch	5	24	0.250	0.191	0.467	0.000
sensor conflict	CVTB-MPC	5	24	0.250	0.166	0.413	0.292
sensor conflict	CVTB no-probe	5	24	0.250	0.166	0.413	0.292
sensor conflict	Oracle	5	24	0.500	0.075	0.467	0.000
sensor conflict	Mean fusion	6	24	0.208	0.185	0.437	0.000
sensor conflict	Ensemble uncertainty	6	24	0.125	0.196	0.429	0.000
sensor conflict	Robust minimax	6	24	0.125	0.191	0.457	0.000

Split	Method	Seed	Ep.	Succ.	Err.	Eng.	Fb.
sensor conflict	Old VT branch	6	24	0.250	0.180	0.495	0.000
sensor conflict	CVTB-MPC	6	24	0.125	0.196	0.427	0.167
sensor conflict	CVTB no-probe	6	24	0.125	0.196	0.427	0.167
sensor conflict	Oracle	6	24	0.625	0.071	0.492	0.000
sensor conflict	Mean fusion	7	24	0.292	0.153	0.433	0.000
sensor conflict	Ensemble uncertainty	7	24	0.250	0.158	0.430	0.000
sensor conflict	Robust minimax	7	24	0.250	0.153	0.442	0.000
sensor conflict	Old VT branch	7	24	0.333	0.148	0.491	0.000
sensor conflict	CVTB-MPC	7	24	0.208	0.162	0.428	0.417
sensor conflict	CVTB no-probe	7	24	0.208	0.162	0.428	0.417
sensor conflict	Oracle	7	24	0.625	0.067	0.462	0.000
sticky contact	Mean fusion	0	24	1.000	0.035	0.681	0.000
sticky contact	Ensemble uncertainty	0	24	1.000	0.034	0.681	0.000
sticky contact	Robust minimax	0	24	1.000	0.021	0.725	0.000
sticky contact	Old VT branch	0	24	1.000	0.033	0.680	0.000
sticky contact	CVTB-MPC	0	24	1.000	0.034	0.680	0.000
sticky contact	CVTB no-probe	0	24	1.000	0.034	0.680	0.000
sticky contact	Oracle	0	24	1.000	0.025	0.763	0.000
sticky contact	Mean fusion	1	24	1.000	0.035	0.681	0.000
sticky contact	Ensemble uncertainty	1	24	1.000	0.035	0.688	0.000
sticky contact	Robust minimax	1	24	1.000	0.022	0.731	0.000
sticky contact	Old VT branch	1	24	1.000	0.035	0.684	0.000
sticky contact	CVTB-MPC	1	24	1.000	0.034	0.688	0.000
sticky contact	CVTB no-probe	1	24	1.000	0.034	0.688	0.000
sticky contact	Oracle	1	24	1.000	0.022	0.765	0.000
sticky contact	Mean fusion	2	24	1.000	0.037	0.679	0.000
sticky contact	Ensemble uncertainty	2	24	1.000	0.036	0.685	0.000
sticky contact	Robust minimax	2	24	1.000	0.022	0.729	0.000
sticky contact	Old VT branch	2	24	0.958	0.038	0.683	0.000
sticky contact	CVTB-MPC	2	24	1.000	0.036	0.678	0.000
sticky contact	CVTB no-probe	2	24	1.000	0.036	0.678	0.000
sticky contact	Oracle	2	24	1.000	0.023	0.777	0.000
sticky contact	Mean fusion	3	24	1.000	0.034	0.671	0.000
sticky contact	Ensemble uncertainty	3	24	1.000	0.033	0.665	0.000
sticky contact	Robust minimax	3	24	1.000	0.019	0.720	0.000
sticky contact	Old VT branch	3	24	1.000	0.033	0.669	0.000
sticky contact	CVTB-MPC	3	24	1.000	0.033	0.670	0.000
sticky contact	CVTB no-probe	3	24	1.000	0.033	0.670	0.000
sticky contact	Oracle	3	24	1.000	0.022	0.755	0.000
sticky contact	Mean fusion	4	24	1.000	0.036	0.675	0.000
sticky contact	Ensemble uncertainty	4	24	1.000	0.035	0.676	0.000
sticky contact	Robust minimax	4	24	1.000	0.023	0.725	0.000
sticky contact	Old VT branch	4	24	1.000	0.036	0.684	0.000
sticky contact	CVTB-MPC	4	24	1.000	0.035	0.682	0.000
sticky contact	CVTB no-probe	4	24	1.000	0.035	0.682	0.000
sticky contact	Oracle	4	24	1.000	0.023	0.761	0.000
sticky contact	Mean fusion	5	24	1.000	0.036	0.670	0.000
sticky contact	Ensemble uncertainty	5	24	1.000	0.035	0.675	0.000
sticky contact	Robust minimax	5	24	1.000	0.021	0.709	0.000
sticky contact	Old VT branch	5	24	1.000	0.035	0.674	0.000
sticky contact	CVTB-MPC	5	24	1.000	0.035	0.671	0.000
sticky contact	CVTB no-probe	5	24	1.000	0.035	0.671	0.000
sticky contact	Oracle	5	24	1.000	0.025	0.763	0.000
sticky contact	Mean fusion	6	24	1.000	0.035	0.657	0.000
sticky contact	Ensemble uncertainty	6	24	1.000	0.035	0.660	0.000
sticky contact	Robust minimax	6	24	1.000	0.021	0.709	0.000
sticky contact	Old VT branch	6	24	1.000	0.034	0.665	0.000
sticky contact	CVTB-MPC	6	24	1.000	0.034	0.660	0.000
sticky contact	CVTB no-probe	6	24	1.000	0.034	0.660	0.000
sticky contact	Oracle	6	24	1.000	0.020	0.742	0.000
sticky contact	Mean fusion	7	24	1.000	0.035	0.680	0.000
sticky contact	Ensemble uncertainty	7	24	1.000	0.034	0.677	0.000
sticky contact	Robust minimax	7	24	1.000	0.023	0.723	0.000
sticky contact	Old VT branch	7	24	1.000	0.034	0.676	0.000
sticky contact	CVTB-MPC	7	24	1.000	0.034	0.680	0.000
sticky contact	CVTB no-probe	7	24	1.000	0.034	0.680	0.000
sticky contact	Oracle	7	24	1.000	0.025	0.755	0.000
tactile noise	Mean fusion	0	24	1.000	0.015	0.422	0.000
tactile noise	Ensemble uncertainty	0	24	0.917	0.026	0.423	0.000
tactile noise	Robust minimax	0	24	1.000	0.039	0.458	0.000
tactile noise	Old VT branch	0	24	0.750	0.049	0.468	0.000
tactile noise	CVTB-MPC	0	24	0.917	0.021	0.420	0.042
tactile noise	CVTB no-probe	0	24	0.917	0.021	0.420	0.042
tactile noise	Oracle	0	24	0.958	0.040	0.397	0.000
tactile noise	Mean fusion	1	24	1.000	0.014	0.411	0.000
tactile noise	Ensemble uncertainty	1	24	0.958	0.020	0.411	0.000
tactile noise	Robust minimax	1	24	1.000	0.036	0.442	0.000
tactile noise	Old VT branch	1	24	0.833	0.036	0.446	0.000
tactile noise	CVTB-MPC	1	24	1.000	0.018	0.407	0.083

Split	Method	Seed	Ep.	Succ.	Err.	Eng.	Fb.
tactile noise	CVTB no-probe	1	24	1.000	0.018	0.407	0.083
tactile noise	Oracle	1	24	1.000	0.038	0.394	0.000
tactile noise	Mean fusion	2	24	1.000	0.015	0.414	0.000
tactile noise	Ensemble uncertainty	2	24	0.958	0.019	0.417	0.000
tactile noise	Robust minimax	2	24	1.000	0.043	0.453	0.000
tactile noise	Old VT branch	2	24	0.917	0.037	0.465	0.000
tactile noise	CVTB-MPC	2	24	0.958	0.015	0.415	0.000
tactile noise	CVTB no-probe	2	24	0.958	0.015	0.415	0.000
tactile noise	Oracle	2	24	0.958	0.040	0.400	0.000
tactile noise	Mean fusion	3	24	1.000	0.013	0.424	0.000
tactile noise	Ensemble uncertainty	3	24	1.000	0.018	0.422	0.000
tactile noise	Robust minimax	3	24	1.000	0.036	0.457	0.000
tactile noise	Old VT branch	3	24	0.875	0.030	0.475	0.000
tactile noise	CVTB-MPC	3	24	1.000	0.017	0.423	0.125
tactile noise	CVTB no-probe	3	24	1.000	0.017	0.423	0.125
tactile noise	Oracle	3	24	1.000	0.039	0.408	0.000
tactile noise	Mean fusion	4	24	1.000	0.013	0.433	0.000
tactile noise	Ensemble uncertainty	4	24	1.000	0.018	0.434	0.000
tactile noise	Robust minimax	4	24	1.000	0.037	0.472	0.000
tactile noise	Old VT branch	4	24	0.792	0.035	0.484	0.000
tactile noise	CVTB-MPC	4	24	1.000	0.018	0.431	0.167
tactile noise	CVTB no-probe	4	24	1.000	0.018	0.431	0.167
tactile noise	Oracle	4	24	0.958	0.040	0.415	0.000
tactile noise	Mean fusion	5	24	1.000	0.013	0.427	0.000
tactile noise	Ensemble uncertainty	5	24	1.000	0.017	0.428	0.000
tactile noise	Robust minimax	5	24	1.000	0.040	0.462	0.000
tactile noise	Old VT branch	5	24	0.917	0.029	0.463	0.000
tactile noise	CVTB-MPC	5	24	1.000	0.017	0.428	0.125
tactile noise	CVTB no-probe	5	24	1.000	0.017	0.428	0.125
tactile noise	Oracle	5	24	1.000	0.038	0.406	0.000
tactile noise	Mean fusion	6	24	0.958	0.018	0.432	0.000
tactile noise	Ensemble uncertainty	6	24	0.917	0.026	0.431	0.000
tactile noise	Robust minimax	6	24	1.000	0.042	0.465	0.000
tactile noise	Old VT branch	6	24	0.750	0.047	0.492	0.000
tactile noise	CVTB-MPC	6	24	0.917	0.022	0.431	0.042
tactile noise	CVTB no-probe	6	24	0.917	0.022	0.431	0.042
tactile noise	Oracle	6	24	0.917	0.038	0.409	0.000
tactile noise	Mean fusion	7	24	1.000	0.015	0.431	0.000
tactile noise	Ensemble uncertainty	7	24	0.958	0.022	0.427	0.000
tactile noise	Robust minimax	7	24	1.000	0.039	0.464	0.000
tactile noise	Old VT branch	7	24	0.875	0.035	0.515	0.000
tactile noise	CVTB-MPC	7	24	1.000	0.020	0.430	0.167
tactile noise	CVTB no-probe	7	24	1.000	0.020	0.430	0.167
tactile noise	Oracle	7	24	0.958	0.038	0.398	0.000
vision bias	Mean fusion	0	24	0.458	0.083	0.390	0.000
vision bias	Ensemble uncertainty	0	24	0.458	0.083	0.389	0.000
vision bias	Robust minimax	0	24	0.375	0.091	0.420	0.000
vision bias	Old VT branch	0	24	0.417	0.084	0.417	0.000
vision bias	CVTB-MPC	0	24	0.417	0.084	0.395	0.000
vision bias	CVTB no-probe	0	24	0.417	0.084	0.395	0.000
vision bias	Oracle	0	24	0.833	0.048	0.393	0.000
vision bias	Mean fusion	1	24	0.417	0.093	0.399	0.000
vision bias	Ensemble uncertainty	1	24	0.417	0.092	0.393	0.000
vision bias	Robust minimax	1	24	0.375	0.104	0.429	0.000
vision bias	Old VT branch	1	24	0.375	0.092	0.432	0.000
vision bias	CVTB-MPC	1	24	0.417	0.092	0.396	0.042
vision bias	CVTB no-probe	1	24	0.417	0.092	0.396	0.042
vision bias	Oracle	1	24	0.750	0.055	0.407	0.000
vision bias	Mean fusion	2	24	0.667	0.068	0.393	0.000
vision bias	Ensemble uncertainty	2	24	0.667	0.067	0.385	0.000
vision bias	Robust minimax	2	24	0.667	0.084	0.423	0.000
vision bias	Old VT branch	2	24	0.667	0.068	0.420	0.000
vision bias	CVTB-MPC	2	24	0.667	0.067	0.390	0.083
vision bias	CVTB no-probe	2	24	0.667	0.067	0.390	0.083
vision bias	Oracle	2	24	0.917	0.045	0.398	0.000
vision bias	Mean fusion	3	24	0.292	0.107	0.389	0.000
vision bias	Ensemble uncertainty	3	24	0.292	0.106	0.386	0.000
vision bias	Robust minimax	3	24	0.250	0.116	0.414	0.000
vision bias	Old VT branch	3	24	0.292	0.109	0.397	0.000
vision bias	CVTB-MPC	3	24	0.292	0.107	0.389	0.042
vision bias	CVTB no-probe	3	24	0.292	0.107	0.389	0.042
vision bias	Oracle	3	24	0.583	0.065	0.409	0.000
vision bias	Mean fusion	4	24	0.542	0.075	0.405	0.000
vision bias	Ensemble uncertainty	4	24	0.542	0.076	0.404	0.000
vision bias	Robust minimax	4	24	0.542	0.087	0.437	0.000
vision bias	Old VT branch	4	24	0.542	0.076	0.433	0.000
vision bias	CVTB-MPC	4	24	0.542	0.076	0.405	0.042
vision bias	CVTB no-probe	4	24	0.542	0.076	0.405	0.042
vision bias	Oracle	4	24	0.917	0.049	0.398	0.000

Split	Method	Seed	Ep.	Succ.	Err.	Eng.	Fb.
vision bias	Mean fusion	5	24	0.542	0.072	0.397	0.000
vision bias	Ensemble uncertainty	5	24	0.542	0.073	0.397	0.000
vision bias	Robust minimax	5	24	0.542	0.087	0.430	0.000
vision bias	Old VT branch	5	24	0.542	0.073	0.429	0.000
vision bias	CVTB-MPC	5	24	0.542	0.073	0.396	0.000
vision bias	CVTB no-probe	5	24	0.542	0.073	0.396	0.000
vision bias	Oracle	5	24	0.833	0.048	0.395	0.000
vision bias	Mean fusion	6	24	0.333	0.117	0.393	0.000
vision bias	Ensemble uncertainty	6	24	0.333	0.118	0.393	0.000
vision bias	Robust minimax	6	24	0.250	0.117	0.426	0.000
vision bias	Old VT branch	6	24	0.333	0.115	0.406	0.000
vision bias	CVTB-MPC	6	24	0.333	0.118	0.393	0.083
vision bias	CVTB no-probe	6	24	0.333	0.118	0.393	0.083
vision bias	Oracle	6	24	0.792	0.063	0.406	0.000
vision bias	Mean fusion	7	24	0.542	0.085	0.398	0.000
vision bias	Ensemble uncertainty	7	24	0.542	0.085	0.399	0.000
vision bias	Robust minimax	7	24	0.500	0.098	0.427	0.000
vision bias	Old VT branch	7	24	0.542	0.085	0.430	0.000
vision bias	CVTB-MPC	7	24	0.542	0.085	0.398	0.083
vision bias	CVTB no-probe	7	24	0.542	0.085	0.398	0.083
vision bias	Oracle	7	24	0.833	0.047	0.402	0.000

H HOSTILE REVIEWER QUESTIONS

Does this prove visuotactile branching is impossible? No. It proves that this implementation and evidence package do not support an ICLR-main positive claim. Better tactile histories, real sensor calibration, richer actions, or learned health likelihoods may revive the broader idea.

Why not submit as a negative result? Negative results can be valuable, but this package lacks real-robot or public benchmark validation. It is submission-grade as an internal archive report and a falsification artifact. That is different from being a high-odds main-conference submission.

Could more tuning make CVTB-MPC win? Maybe on this custom benchmark, but post-freeze tuning would be scientifically weak. A new method should receive a new development log and frozen evaluation.

Why include the oracle? The oracle is not a fair deployable baseline. It measures remaining headroom if hidden mode information were available. An oracle gap shows the task has avoidable failures; it does not validate this method.

Why insist on a 25+ page PDF for an archive? Because the expanded standard is about evidence quality, not optimism. A failed paper should fail with enough formal setup, ablations, diagnostics, and reproducibility detail that the next attempt knows exactly what not to repeat.

I TERMINAL DECISION

The final decision is KILL-ARCHIVE. The code, frozen protocol, generated tables, validation script, and final PDF remain useful for future work. The current mechanism should not be sent to ICLR main as a positive contribution unless a future rebuild changes the method and survives a new frozen protocol.

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